Revisiting Economic Growth: Frictions, Complexity and Productivity

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Abstract

We study whether economic complexity helps explain productivity differences, and why complex production emerges unevenly across countries. Cross-country development accounting and panel regressions show that the Economic Complexity Index is robustly associated with higher TFP levels and predicts TFP growth. We microfound this link in a multi-industry Melitz framework where fixed production and export costs are log-supermodular in product complexity and country-level frictions. Manufacturing-export evidence shows that institutional and logistical frictions disproportionately reduce export activation and export values for complex products. Development requires both productive capabilities and an institutional environment that allows them to operate.

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1 Introduction

"Our understanding of economic growth is less definitive than commonly thought—even though advice sometimes has been given to developing countries with greater confidence than perhaps the state of our knowledge would justify."

- World Bank Commission on Growth and Development (2010)

The question of the fundamental causes of differences in income between countries is as essential as it is contentious. The history of the literature on this topic can be summed up by a gradual descent down the objects of the neoclassical production function first introduced by Solow (1956) & Swan (1956). Capital accumulation, labour force participation, and human capital all momentarily enjoyed their place in the sun as the primary determinant of income differentials. The determinants of national income are therefore disaggregated into (i) factor endowments (K, L) and later H and (ii) a residual "A", representing Total Factor Productivity (TFP), or the efficiency with which an economy uses its factor endowments. This residual has come to encompass technology, institutions, and culture, among other intangible determinants of growth.

The central implication of the neoclassical model is that, under diminishing returns to capital, capital accumulation alone cannot sustain long-run growth in income per worker. In the long run, sustained growth must come from improvements in productivity. Moreover, growing levels of education, access to healthcare, and the globalisation of capital flows between countries have made many of these factors relatively abundant in different parts of the world (Barro and Lee, 2013; Scott, 2023; UNCTAD, 2022), but widespread growth remains as elusive as ever. In developed and developing countries alike, the question of how to generate continuous, sustainable growth is at the forefront of policy debates.

Much of the modern growth literature can be read as an attempt to uncover what is inside TFP. Endogenous growth theory emphasizes innovation, knowledge accumulation, and technological change (Romer, 1990; Aghion and Howitt, 1992). A related development literature stresses that productivity also depends on deeper determinants, such as culture and, especially, institutions, which shape incentives, property rights, contract enforcement, and, overall, the allocation of resources across the economy (Acemoglu et al., 2001, 2005).

We argue that an important and overlooked component of the Solow residual is *productive capabilities*. Productive capabilities are the latent ability to seamlessly combine complementary factors through various production processes. Although they are latent and cannot be directly observed, they can be inferred from the set of final goods a country produces. This realisation of a country's productive capabilities is referred to as *economic complexity*. Countries which produce diverse and sophisticated sets of products are said to be more complex, and must necessarily possess more advanced productive capabilities than others (Costinot, 2009b). This means they utilise their factor endowments more efficiently in the present and possess potential for innovation in the future.

Our research contributes to the literature on the fundamental drivers of growth by exploring the role of economic complexity, measured by Harvard Growth Lab's Economic Complexity Index (ECI), in explaining differences in Total Factor Productivity (TFP) and income. This paper is divided into two parts. In the first part of our analysis, we find that there is a robust, positive relationship between ECI and TFP in levels, and that ECI today is a powerful predictor of future TFP growth, concluding that ECI may explain up to 13% of differences in income between countries and $\sim$28% of cross-country variation in TFP as a lower bound. We justify our findings by appealing to literature on linkages, knowledge spillovers, and searching-and-matching dynamics, which helps explain why an economy with a diverse set of sophisticated industries is more productive on the aggregate.

The second part addresses the more fundamental question of the drivers of differences in economic complexity between countries and across time. We develop a simple model where the presence of institutional and logistical

frictions discourages firms from entering into more complex sectors. The core property of this model is that the fixed costs of production and exporting are log-supermodular in product complexity and country-level frictions. Empirically we find indications that a high friction environment disproportionately harms firms producing more complex products: for exports, a one-standard-deviation increase in frictions is associated with a differential decline of between 5% and 14% for products that are one standard deviation more complex. For export activation, a one-standard-deviation increase in frictions is associated with a differential reduction in the probability of exporting of between 0.4 and 0.9 percentage points. In relative terms, this implies an added penalty of roughly 7–11% over the estimated direct effect of frictions.

While an increase in factor accumulation is a necessary condition for long-run growth, we argue that it is insufficient, and must be accompanied by entry into increasingly more complex sectors. This allows countries to benefit from numerous reinforcing spillovers, linkages and externalities within and between sectors. Frictions serve as a barrier against this virtuous cycle, and the elimination of frictions presents itself as a powerful mechanism for enabling sustainable productivity growth through the channel we identify.

The remainder of the paper is organised as follows. Section 2 reviews the theoretical and empirical literature on Economic Complexity. Section 3 outlines three stylised facts linking ECI to TFP using cross-country accounting exercises and explores the underlying mechanism behind these stylised facts. Section 4 links complexity with firm-level frictions. Section 5 presents our formal theoretical model on firm entry into complex sectors. Section 6 describes our data and empirical strategy, while Section 7 presents our results. Section 8 concludes.

2 Literature Review: Complexity in Theory and Empirics

A relevant insight of recent literature is that the composition of production contains information about productivity. Countries differ not only in their levels of accumulated factors of production, but also in productive capabilities which are reflected in the goods they produce.

While this intuition traces back to classical arguments regarding manufacturing diversity and structural linkages (Hamilton, 1791; List, 1841; Hirschman, 1958), it has been modernised by a literature analysing product complexity (Hidalgo and Hausmann, 2009; Hausmann and Hidalgo, 2022). The claim is not that a country's export basket mechanically causes development. Rather, the sophistication of the export basket reveals latent productive capabilities: the economy's systemic capacity to combine complementary inputs across production processes. These capabilities matter because they dictate how an economy can upgrade, diversify and transition to higher-productivity activities that sustain long-term growth.

The most intuitive case for this argument comes from Kremer's O-ring model. Kremer (1993) proposes that producing a good p requires N_p tasks, and each task is completed successfully with probability q_i. Thus, the probability of successfully producing the final good is then:

$$P_p = \prod_{i=1}^{N_p} q_i.$$

If we assume all tasks to have the same success probability q, this becomes:

$$P_p = q^{N_p}.$$

In this setting *complexity* is defined by probability of successfully producing a good, which makes it an increasing function of the number of complementary tasks that must be combined to produce a certain good. Kremer illustrates this distinction by comparing relatively simple goods, such as textiles, with more complex goods, such as aircrafts.

As the number of required tasks rises, there are more points at which production can fail. The coordination burden therefore increases with complexity: more tasks must be completed reliably for production to be viable. As a result, Kremer posits that rich countries specialise in complex goods due to having a higher average task reliability across all tasks. In turn, this allows them to coordinate production processes across many complementary tasks. Conceptually, this property of average task reliability is what we refer to as *productive capabilities.*

Costinot (2009b) embeds these notions of complexity and capabilities in the context of international trade. He defines a good's complexity as the number of tasks required to produce one unit. Because tasks are complementary, countries that can perform tasks more reliably (due to better institutions and human capital) hold a comparative advantage that is increasing in a good's complexity. Complexity and the capacity to execute reliably are therefore complementary. For more developed economies, this implies that their advantage lies not only in having greater factor endowments, but in being able to coordinate those factors across more demanding production processes.

Productive capabilities also have long-run dynamic effects. As shown by Atkin et al. (2025), what a country produces today shapes the capabilities it has tomorrow, and producing complex goods can build the capacity to produce still more complex goods in the future. The implication is a self-reinforcing process that drives growth, since frontier technologies serve as a foundation for future innovation.

Taken together, these papers give complexity a simple theoretical meaning. Products require sequences of tasks; countries with greater productive capabilities can perform a broader set of tasks reliably, developing the capacity to combine complementary inputs across diverse production processes, and may therefore be able to produce more complex products.

A more complex good is not intrinsically "better", it merely requires a larger set of complementary tasks to be performed well. Producing socks requires relatively few tasks. A semiconductor requires many interdependent high-precision tasks: intricate chip design, multiple steps of precision fabrication, sourcing of specialised inputs and equipment, complicated logistics, finance, quality control, contract enforcement and so on. Only countries that can reliably perform this fuller set of tasks can export the more complex good.

The difficulty is that the number of tasks required for producing any particular good, and by extension productive capabilities countries possess are both latent properties. [1] They cannot be observed directly across the full set of countries and years required for cross-country work. Kremer (1993) suggests that one possible measure of complexity is the number of different inputs required in production. However, there was no subsequent convergence on specific methods to measure or proxy for latent productive capabilities empirically.

Instead, a separate empirical literature emerged in this vacuum. A seminal paper by Hidalgo and Hausmann (2009) introduced the ECI which infers a country's productive capabilities ex-post through its export basket. According to the ECI, countries that export many different products are likely to possess a broader set of capabilities. Products exported by few countries are likely to require more specialised capabilities. Economies that combine high export diversity with low product ubiquity are therefore classified as more complex.

ECI has proven to be a powerful tool for predicting major macroeconomic trends, from serving as a robust predictor of economic growth (Hausmann and Hidalgo, 2022) to long-run changes in income inequality (Hartmann et al., 2017) to a country's future industrial carbon emissions (Romero and Gramkow, 2021).

It is important to note that ECI can only approximate productive capabilities, as they are inherently unmeasurable. ECI defines complexity deductively by reading off the components of a country's export basket. In this respect, ECI is closer to Costinot's trade setting: countries that can reliably perform more tasks should reveal

[1] Kremer provides an instructive example in his original paper: "[M]aking a car involves more tasks than making a bicycle, because there are more things that can go wrong. It is more difficult to make cardinal statements about the number of tasks. For example, a waiter can be thought of as performing a single task [...] or three tasks-taking the order, serving the food, and collecting the check."(Kremer, 1993, p.11.)

this ability by exporting more complex goods. The ECI is thus informative about the productive capabilities implied by the task-based theory. It remains, however, an imperfect proxy rather than a direct count of the number of tasks required to produce a good.

In this respect, the theoretical and empirical literature are largely disjointed. Moreover, the ideas developed in the empirical complexity literature have not been fully integrated into the broader growth and development literature. Atkin et al. (2025) is an important exception. They explicitly connect a task-based theory of capabilities to the empirical complexity literature, showing how the evolution of production structure can be studied through an ECI-style measure. Our paper is related in spirit.

Our contribution is twofold. In the first part, we explicitly link complexity to the standard cross-country growth accounting framework. In doing so, we provide some motivating facts and quantify the potential impact of complexity on TFP. In the second part, we propose theoretically and explore empirically a channel through which complexity emerges due to complementarities between country-level frictions and complex sectors.

We begin by exploring three motivating facts that link Complexity to TFP in a cross-country setting.

3 Motivating Facts

3.1 Data

In the first part of our empirical analysis we ask whether economic complexity can help account for part of the cross country differences in TFP. [2] This focus reflects the central role of productivity gaps in explaining cross-country income dispersion (Caselli, 2005, 2016; Jones, 2016). In doing so, we follow a long tradition of seeking to reduce what Abramovitz (1956) called our "measure of ignorance."

We use ECI to capture cross-country differences in economic complexity. Our headline measure is the SITC-based Economic Complexity Index from The Growth Lab at Harvard University (2026b). While the literature on complexity has developed a variety of alternative methodologies for measuring complexity, the measure we choose is appropriate for our purposes because it provides the longest available panel, beginning in 1970 for our sample, and remains the benchmark against which alternative measures are assessed. Our main estimation sample contains 124 countries.

In practice, ECI is a standardised relative index centred around zero. It is bounded between -3 to 3 across country-year observations, with lower values indicating less complex countries. For example, a country with an ECI of 2 is not “twice as complex” as a country with an ECI of 1. Rather, it occupies a higher relative position in the global distribution of revealed productive capabilities. A more detailed discussion of ECI is provided in Appendix A.1.

An enduring concern in the growth and development literature is that, while many variables have been claimed to matter empirically for growth, it is often unclear which relationships are quantitatively robust (Levine and Renelt, 1992; Sala-i Martin, 1997). This motivates our inclusion of a broad set of standard growth controls. Some controls capture deep determinants of development. For example, institutions[3] shape incentives to invest, innovate and accumulate capabilities, while geography may affect development through climate, disease burdens, and access to markets (Hall and Jones, 1999; Gallup et al., 1999). Other controls capture macroeconomic channels, including investment, trade openness, government expenditure, demographics and inflation (Barro, 1991; Mankiw et al., 1992). Although we do not interpret our estimates causally, these

[2] A detailed description of how we construct TFP is included in Appendix A.2.

[3] Here it is appropriate to be specific about what we mean by institutions. We appeal to the broader notion set out by North, who defines institutions as “humanly devised constraints that shape human interaction” North (1990). We focus exclusively on the fact that Institutions may meaningfully shape how “people, organisations and firms utilise resources in political and economic decision making” (Sharma (2007), p. 33).

variables help assess whether the complexity-TFP relationship is distinct from other commonly used correlates of development. [4][5]

We control for natural resource (NR) rents throughout, as ignoring natural capital can bias residual measures of productivity, as part of the Solow residual may reflect oil, gas, mineral or forestry rents rather than actual efficiency (Brandt et al., 2017; Caselli and Feyrer, 2007).

3.2 Regressions

We estimate two sets of regressions. The first examines whether more complex economies have higher TFP levels. The second asks whether initial complexity predicts subsequent ten-year TFP growth. Given longstanding scepticism around the robustness of cross-country growth regressions, we assess whether these relationships are robust to alternative conditioning sets.

We first estimate the relationship between economic complexity and the level of TFP:

$$\log A_{i,t} = \alpha + \beta_{level} \, \mathrm{ECI}_{i,t} + X'_{i,t}\gamma + \delta_t + \varepsilon_{i,t}. \tag{1}$$

Here, $\log A_{i,t}$ is the log TFP residual for country i in year t, $\mathrm{ECI}_{i,t}$ is the Economic Complexity Index, $X_{i,t}$ is a vector of controls, δ_t denotes year fixed effects, and $\varepsilon_{i,t}$ is the error term. Standard errors are clustered at the country level. The coefficient of interest is β_{level}, which captures the partial correlation between economic complexity and productivity.

We then estimate whether initial complexity might predict subsequent TFP growth:

$$\Delta \log A_{i,t} = \alpha + \beta_{growth} \, \mathrm{ECI}_{i,t-10} + \phi \log A_{i,t-10} + X'_{i,t-10}\gamma + \delta_{t-10} + \varepsilon_{i,t}. \tag{2}$$

The dependent variable is the ten-year change in log TFP, defined as

$$\Delta \log A_{i,t} = \log A_{i,t} - \log A_{i,t-10}.$$

All right-hand-side variables are measured at the beginning of the period. Initial $\log A_{i,t-10}$ captures convergence effects. Start-year fixed effects absorb common initial conditions (Islam, 1995).

In this setting, the coefficient β_{growth} measures whether initial complexity predicts subsequent TFP growth, conditional on convergence (and the included controls). Following Hidalgo and Hausmann (2009), we use lagged rather than contemporaneous ECI because the relevant hypothesis is that existing productive capabilities predict subsequent growth.

For these two regressions, the baseline specification controls for natural resource rents. The intermediate specification adds standard macroeconomic and structural controls. The full specification adds institutional quality[6], a landlocked dummy, and absolute latitude, capturing deeper institutional and geographic correlates of development. This ordering shows whether the ECI coefficient survives standard growth controls before introducing the deeper determinants emphasised in the development literature.

Despite our aim to establish a robust positive correlation between complexity and TFP, four concerns prevent a causal interpretation of our results.

The first limitation is reverse causality. More productive countries may be better able to produce and export complex goods, so causality may run from TFP to ECI. The second is the risk of omitted variable bias. Despite our efforts, there are likely to be omitted determinants of TFP that are correlated with ECI. Third, and arguably

[4]The main analysis includes a non-exhaustive set of control variables. Including all possible determinants of growth in a single regression would be infeasible. We provide robustness checks using a more extensive set of potential controls in Appendix A.4.

[5]Descriptive statistics and further discussion of the data used for these controls is provided in Appendix A.1.

[6]Specific details on the construction of this control, derived from WGI indicators on institutional quality, are provided in 6

more important than omitted variables, some of our controls may lie on the causal pathway (Cinelli et al., 2024). Institutions are the clearest example. They may raise productivity directly, but they may also shape the incentives that allow complex productive structures to emerge. Finally, despite the theoretical link between productive capabilities and ECI, our results are subject to measurement error, since ECI is an imperfect proxy for latent productive capabilities. Together, these concerns could bias the estimates in either direction.[7]

A possible solution would be to instrument for ECI, but in this setting the exclusion restriction is extremely demanding, and the literature on complexity has not, to our knowledge, established a credible instrument for our purposes. Given these limitations, the main objective of our regression is to show that ECI remains significant and positive across all specifications; we follow the cross-country growth literature in treating our estimates as conditional correlations rather than identified causal effects (Caselli, 2005; Hausmann et al., 2014). The next section lists our findings as stylised facts to motivate the theoretical analysis.

3.3 Motivating Facts

In this section, we document several patterns from our cross-country regressions that investigate the link between ECI and TFP at the macro level. These patterns validate our central hypothesis and motivate our subsequent modelling and investigation into what might inhibit the emergence of firms operating in a complex sector given a particular country.

First, more complex countries exhibit higher TFP in levels.

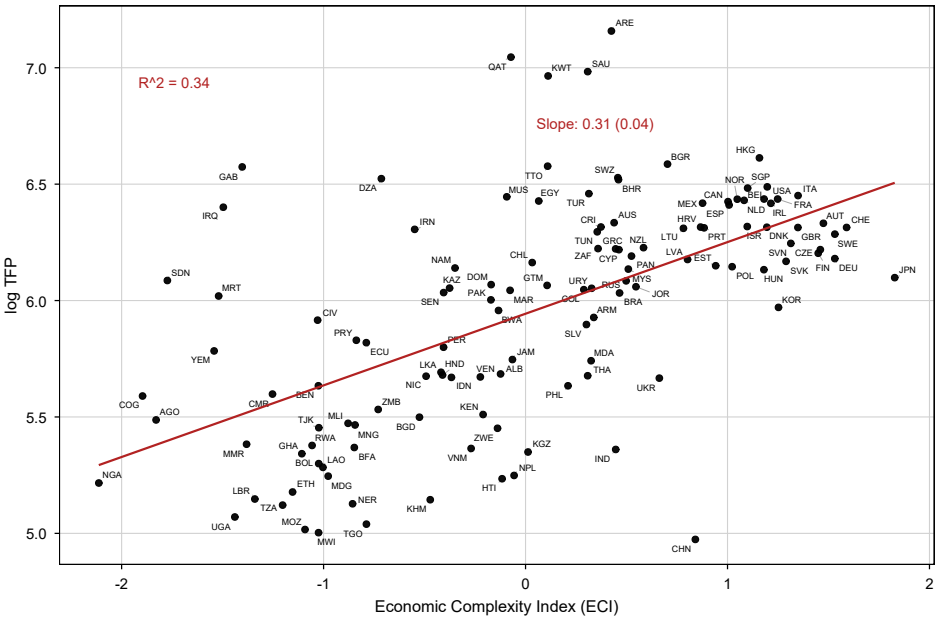


Figure 1: Economic complexity and total factor productivity across countries. Each point represents one country, plotted by its sample-average ECI against its sample-average log TFP.

Table 1 reports the relevant regressions of TFP and TFP-growth on ECI. Column (1) shows that there is a strong correlation between ECI and log TFP when controlling for NR Rents. Figure 1 shows this relationship visually.[8] Column (2) adds macroeconomic and geographic controls. The coefficient on ECI remains stable despite the addition of these controls. This suggests that the correlation is not merely driven by established macroeconomic relationships such as investment share or export share of GDP, nor can it be attributed to

[7] In sum, these concerns are common throughout the broader growth and development literature. For example, Hanushek and Woessmann (2020) discuss the difficulty of identifying causal effects in the relationship between education, knowledge, and economic growth.

[8] Unless stated otherwise, the fitted line is the OLS regression of log TFP on ECI; the reported slope (standard error in parentheses) and R^2 summarise the cross-country association.

traditional explanations such as geographic endowments. Complexity is plausibly a new channel that captures the effect of productive capabilities on TFP.

On the other hand, a large body of work argues that institutions are a (if not *the*) fundamental determinant of long-run development, shaping incentives to invest, innovate and allocate resources towards productive activities (North, 1990; Acemoglu et al., 2001, 2005; Acemoglu and Robinson, 2019). This matters for our setting because institutions may affect both sides of the relationship: they can raise aggregate productivity directly, while also supporting the accumulation of productive capabilities. Vu (2022) argues that better institutions may foster economic complexity by strengthening incentives for innovative entrepreneurship, supporting human capital accumulation, and directing resources towards more productive activities.

Here we control for institutional quality as a regressor in column (3). Including institutional quality reduces the ECI coefficient by roughly half, from 0.330 in column (2) to 0.184 in column (3). Nevertheless, the ECI coefficient remains positive and statistically significant. This attenuation suggests that part of the ECI-TFP correlation is shared with institutional quality. However, the regression does not tell us whether institutions shape complexity, complexity shapes institutions, or if both reflect a deeper development process. Full regression tables and comprehensive robustness checks for alternative constructions of TFP and regional effect dummies are provided in Appendix A.5.

Table 1: Economic Complexity and Total Factor Productivity

	TFP levels: $\log A_{i,t}$			Ten-year TFP growth: $\Delta \log A_{i,t}$		
	Base	+ Macro & Geography	+ Institutions	Base	Macro & Geography	+ Institutions
	(1)	(2)	(3)	(4)	(5)	(6)
ECI / Initial ECI	0.384***	0.330***	0.184**	0.129***	0.128***	0.102***
	(0.035)	(0.034)	(0.056)	(0.020)	(0.025)	(0.032)
Initial TFP, $\log A_{i,t-10}$	–	–	–	−0.291***	−0.330***	−0.407***
				(0.035)	(0.048)	(0.086)
Controls included:						
NR rent controls	Yes	Yes	Yes	Yes	Yes	Yes
Macro controls[a]	No	Yes	Yes	No	Yes	Yes
Geography controls[b]	No	Yes	Yes	No	Yes	Yes
Institutional controls[c]	No	No	Yes	No	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SEs	Country	Country	Country	Country	Country	Country
Observations[d]	5,472	5,063	2,745	4,491	4,095	1,777
R^2	0.400	0.518	0.606	0.248	0.267	0.262
Within R^2	0.384	0.505	0.597	0.166	0.173	0.183

[a] Macro controls: Investment/GDP, Inflation, Log population, Exports/GDP, Government/GDP.
[b] Geography controls: Landlocked (all columns except 1,4) and Absolute latitude (only column 3).
[c] Institutional control: The first principal component of the six World Governance Indicators; details in section 6.1.
[d] WGI data begins in 1996, eliminating ∼half of the observations when controlling for institutions.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Standard errors clustered at the country level in parentheses. Statistical significance is reported for correlational estimates only and does not imply causation.

Second, countries with higher complexity tend to have higher subsequent TFP growth.

Columns (4)–(6) ask a distinct but related question: whether complexity predicts future productivity growth, not just higher productivity in levels. Positive and statistically significant coefficients provide support for the hypothesis that ECI is associated with subsequent TFP growth. Mirroring the levels regressions, the coefficient changes only modestly with the addition of macroeconomic and geographical controls between columns (4) and (5), but falls noticeably after controlling for institutions in column (6).[9]

[9]Interestingly, complexity works against cross-country TFP convergence, as seen by the relatively large negative coefficient on the initial TFP regressor compared to the positive coefficient on lagged ECI.

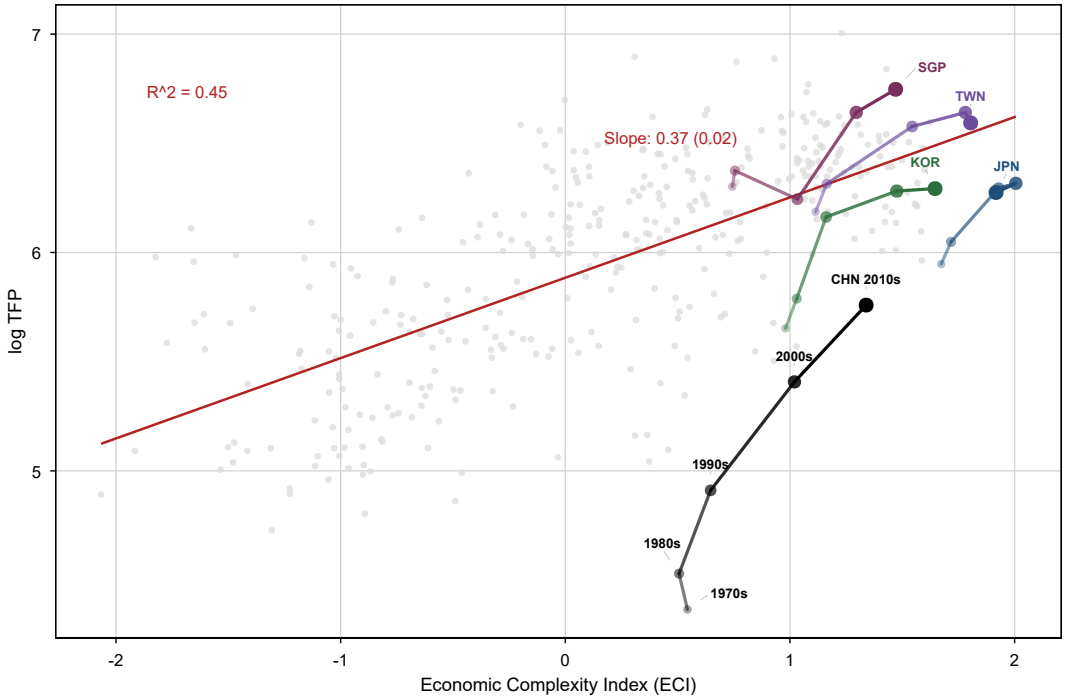


Figure 2: Economic complexity and log TFP across country-decade observations. Each grey point is one country-decade average; the fitted line is a pooled regression across all observations, summarising the broad cross-country association. Coloured paths trace selected East Asian economies (China, Japan, Korea, Singapore, Taiwan) across decades.

The coloured paths in Figure 2 trace several East Asian economies that moved up the complexity distribution over this period. China provides an illustrative case that is at least partially consistent with this observation. As Rodrik (2006) argues, an important determinant of China's rapid growth was the relative sophistication of its export basket compared to its income level. Through the lens of complexity, this suggests that China's production structure embodied capabilities more advanced than its income level alone would imply. Its subsequent growth is therefore consistent with the idea that revealed productive capabilities can predict future development.

Third, ECI may explain up to 13% of differences in income between countries, and on average, ∼28% of cross-country variation in TFP.

The first two facts show that the ECI–TFP relationship is positive and robust across specifications. They do not tell us how economically important the relationship is.[10] We therefore use development accounting to quantify how much cross-country variation in complexity matters. To keep our findings as comparable as possible to the previous literature, we follow the accounting approach of Klenow and Rodríguez-Clare (1997), Hall and Jones (1999) and in particular Caselli (2005). For the purpose of our motivating facts, we use the Klenow Rodriguez-Clare variance decomposition. An explanation of this method, along with other standard decompositions and their results is provided in Appendix A.4.

In short, we now ask the question: *Of the variance in income that physical and human capital accumulation alone cannot account for, how much is associated with differences in countries' productive capabilities?* Figure 3 illustrates this using 2016 as the base year, showing how the TFP residual shrinks once complexity is included, and how the share explained by complexity is halved once we control for institutions.

We perform this exercise for the same panel of countries every year between 2002-2023, finding that ECI accounts for a lower bound of 13% of cross-country variance in income. For reference, physical and human

[10] For the reasons discussed above (i.e. reverse causality, omitted-variable bias, and the possibility that some controls lie on the causal pathway) the regression coefficients do not have a clean causal interpretation.

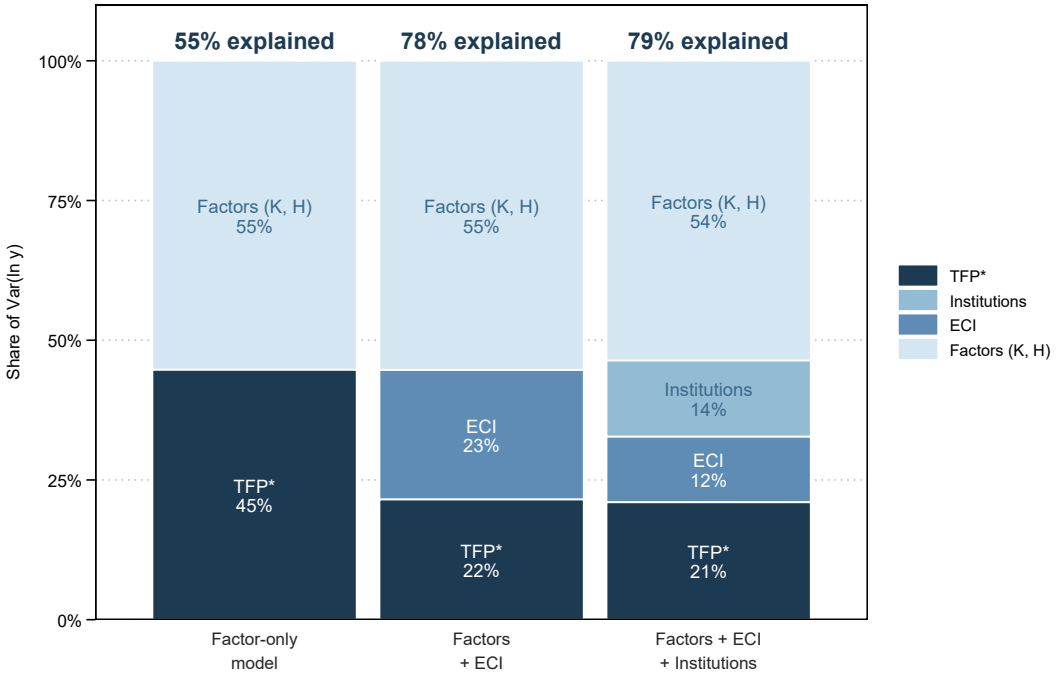


Figure 3: Variance decomposition of cross-country income per worker for 2016, following the covariance-share decomposition of Klenow-Rodriguez-Clare, described in Appendix A.4. The TFP Residual meaningfully shrinks after the introduction of ECI, but adding institutions attenuates the share explained.

capital together account for up to 55% of that variance. Once institutions are added, the contribution of ECI falls substantially, to an upper bound of about 7%, while institutions account for up to roughly 18% of cross-country income variance. A complementary, back-of-the-envelope calculation asks the same question of variation in TFP directly rather than of income.[11] Under this crude approach, ECI accounts for on average 28% of the cross-country variance in TFP in the unrestricted case, and an average of 15% when controlling for institutions.

3.4 Why do we see these facts?

Three broad mechanisms link economic complexity to TFP in levels, three further mechanisms link it to TFP growth specifically.

In levels, a more complex economy could exhibit higher TFP through three channels. First, more capabilities allow countries to produce in a wider set of sectors, generating gains on the extensive margin (Hummels and Klenow, 2005) and forward-backward linkages between sectors (Jones, 2011). This aligns with classic development economists, who argued that structural transformation requires weaving an "interlocking web" of such linkages, with manufacturing acting as the dynamic engine of economy-wide technological spillovers (Hirschman, 1958; Kaldor, 1966). Second, producing complex goods requiring many complementary tasks yields higher-value outputs, since only economies that can reliably coordinate those tasks can export them (Kremer, 1993; Hausmann et al., 2007). Third, borrowing the concept of agglomeration effects from New Economic Geography, the co-location of diverse, complex activities creates "thick" markets that improve worker-firm match quality, as the probability and quality of matches both increase with market size and diversity (Helsley and Strange, 1990; Wheeler, 2001; Duranton and Puga, 2004).

Complexity additionally contributes to TFP growth through three channels. First, complex sectors have high learning elasticities, making learning-by-doing and R&D spillovers most potent in exactly those activities (Greenwald and Stiglitz, 2006). Second, non-rivalrous knowledge spillovers from proximity to high-skill,

[11] This is an intentionally crude calculation, reported only to give an impression of the magnitude of complexity's importance for TFP. We simply compute the share of the cross-country variance of log TFP attributable to ECI in each annual cross-section, using the lower bound of the share of variance attributed to ECI, and divide it by the share of income variation left unexplained by factors.

complex sectors facilitate both accidental and intentional technology transfer (Lucas, 1988; Moretti, 2021). Third, innovation is in part a process of finding new useful combinations of existing processes: a larger set of reliably executable tasks exponentially expands the feasible set of recombinations, and therefore leads to greater potential for innovation (Weitzman, 1998; Fleming and Sorenson, 2001).

China's growth illustrates these mechanisms concretely. Beginning in the 1980s with labour-intensive textiles and simple assembly, manufacturers gradually moved into electronics and higher-precision production as capabilities in logistics, process engineering, and supplier coordination accumulated (Zhu, 2012). The geographic concentration of these activities, most visibly in the Pearl River Delta, created thick labour markets that improved worker-firm matching (Moretti, 2010) and facilitated knowledge spillovers across geographically overlapping supply chains (Smarzynska Javorcik, 2004).

3.5 How can we disentangle complexity and institutions?

As we have repeatedly seen, adding institutions attenuates the independent contribution of ECI to TFP, but does not eliminate it. ECI and institutional quality can be said to capture overlapping dimensions of economic development. This is consistent with the argument that complexity proxies for productive capabilities which may be a deeper determinant of growth, though their effect is difficult to disentangle from other deep determinants. This raises the question: how do institutions shape productive capabilities, and how do those capabilities translate into productivity growth? We depict this relationship as follows[12]:

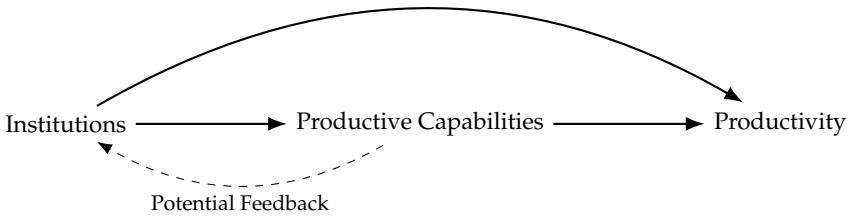


Figure 4: Institutions, productive capabilities, and productivity. Solid arrows denote direct structural channels; the dashed arrow denotes potential feedback from capabilities to institutions.

This relationship between institutions and productivity through the channel of productive capabilities has received limited attention in the growth and development literature. One reason for this separation is that institutions and complexity are often presented as competing explanations of development. We view them instead as complements. For example, institutions have been argued to shape the incentives that determine how talent is used. Where property rights are secure and corruption is limited, skilled individuals have stronger incentives to build firms and enter new activities (Baumol, 1990; Murphy et al., 1991). This can be seen as one channel through which productive capabilities can also accumulate, raising the possibility of higher complexity.

The reverse channel is also plausible. More complex production may create demand for better coordination, stronger contract enforcement, and more capable public institutions. Hartmann et al. (2017) find that economic complexity and inclusive institutions tend to co-evolve, consistent with this narrative of feedback from productive capabilities to institutions.

Our aim is not to definitively disentangle this chain of causation using cross-country data. Instead, we take seriously the view that both institutions and productive capabilities matter, focusing on a mechanism that links them. The question we ask is not whether institutions or complexity fundamentally *cause* growth, but why complex productive sectors emerge in some countries but not in others. The second part of the paper studies one mechanism through which institutional conditions may shape the emergence and performance of complex sectors.

[12]We abstract from direct effects of institutions on productivity, as well as potential feedback from productivity to institutions and productive capabilities.

4 From Macro to Micro

An extensive literature studies how a lack of institutional and logistical quality constrains firms' ability to produce, operate, and access markets.[13] Henceforth, we refer to these constraints as *frictions*. Do and Levchenko (2009) make this link explicit by modelling institutional quality as a fixed cost of production: worse institutions imply higher fixed costs and a smaller mass of active firms.

To make this mechanism explicit, we build on the heterogeneous-firm framework of Melitz (2003). In that setting, fixed costs determine which firms are productive enough to operate and which firms are productive enough to serve foreign markets.

We extend the Melitz framework by allowing these fixed costs to vary with country-level frictions and sector-level complexity, connecting it with the basic theory of complexity from Kremer (1993) and Costinot (2009b). Recalling Kremer (1993)'s O-ring model introduced in Section 2, producing product p requires the successful execution of a chain of N sequential, independent tasks, where the number of tasks is higher for more complex goods ($N(C_p)$).

We assume that the probability of completing a task is affected by the quality of the production environment. Frictions represent a reduction of this quality, such that we can model the probability of successfully executing any individual task as $q(F_c)$, with $q'_{F_c}(F_c) < 0$. Thus the joint probability of completing the entry sequence is $[q(F_c)]^{N(C_p)}$.

Fixed cost can be thought of as the costs a firm needs to incur to create a production environment that *guarantees* a probability of success that allows for profitable production. If increasing frictions now decrease the probability of fully executing each single task by δ, the whole probability of success is reduced by $\delta^{N(C_p)}$. In order to offset that reduction and maintain the same necessary level of probability of success from before, a firm needs to pay additional fixed costs proportional to $\delta^{-N(C_p)}$. From this follows that the fixed costs can be expressed as being directly proportional to the inverse of the probability of success:

$$f^d \propto [q(F_c)]^{-N(C_p)}$$

This allows us to connect frictions and complexity to firm production and activation within an established model of trade (Melitz, 2003). We formalise two propositions supported by the literature:

Proposition 1. *Country-level frictions raise the fixed costs of operating in, or serving, domestic and export markets.*

A broad literature shows that financial constraints, costly firm-entry requirements, and weak court enforcement raise barriers to firm operation, investment, and market access, consistent with the idea that frictions increase the fixed costs firms must incur before production and exporting become viable (Djankov et al., 2002; Manova, 2013; Ponticelli and Alencar, 2016). Logistics frictions provide a further channel. Donaldson (2018) shows that transportation infrastructure reduced trade costs, narrowed price gaps, increased trade, and raised real income in colonial India. In our framework, weak logistics infrastructure raises the resources, the fixed costs, firms must commit to reliably complete tasks and produce. The export margin follows the same logic. Helpman et al. (2008) use entry-regulation measures from Djankov et al. (2002) as proxies for fixed trade costs in an adapted Melitz Model, arguing that these measures affect firm-level fixed rather than variable costs of trade. This makes the paper directly relevant for our mechanism: institutional barriers to market access can raise the fixed cost of serving foreign markets, thereby reducing the set of firms able to export. While these papers point to different frictions, they fit a common definition of frictions as country-level constraints that make production or market access harder to organise reliably. Their shared implication for our framework is that a high-friction

[13] To clarify our definitions, we explore frictions as a broad category, within which political institutions as well as physical infrastructure and logistics affect decision-making at the firm level. Evidence from the literature points towards the two being intimately related as determinants of the overall business environment (Azzimonti, 2011; Busso et al., 2012).

environment raises the resources firms must commit before production can be organised with sufficient legal, financial, logistical, and operational reliability.

These arguments make the case that institutional frictions raise the fixed cost of production. Yet, the central implication of the model is stronger: the fixed-cost penalty from frictions is relatively larger in more complex sectors since the penalty increases in the number of tasks ($\delta^{-N(C_p)}$).

This motivates the second proposition:

Proposition 2. *Complexity raises fixed costs of producing and exporting, and fixed costs are raised relatively more by frictions in more complex sectors.*

In other words, fixed costs increase with product complexity and are log-supermodular in country-level frictions and sector complexity: frictions matter more precisely where production is more complex. This assumption fits naturally within the trade literature on institutional comparative advantage. Costinot (2009a) provides the logic for the effect of log-supermodularity on comparative advantage: country characteristics matter most in sectors that use the corresponding capability intensively. The institutional trade literature has the same interaction structure. Weak contract enforcement, contractual incompleteness, and insecure property rights do not affect all sectors equally; they impose larger distortions in sectors that rely more heavily on differentiated goods, relationship-specific inputs, complementary intermediates, and coordination among many parties (Berkowitz et al., 2006; Ranjan and Lee, 2007; Acemoglu et al., 2007; Levchenko, 2007; Nunn, 2007).

Although these papers do not measure complexity in the same way as we do, they identify the production features that make products institutionally demanding. Complex products are more likely to require exactly these features: complementarity, coordination, relationship-specific investment, reliable contracting, and sophisticated logistics. The empirical implication is therefore an interaction between country-level frictions and product-level complexity. Holding constant country-year, product-year, and country-product conditions, frictions should reduce performance relatively more for complex products than for less complex products. This prediction guides the product-level analysis below.

5 Theoretical framework: Product-indexed Melitz industries

The theoretical framework builds on the heterogeneous-firm mechanism of Melitz (2003). In the spirit of multi-industry heterogeneous-firm models such as Bernard et al. (2007) we move from a one-industry case to a case where each product p is interpreted as a differentiated product industry populated by a continuum of heterogeneous firms. Each firm produces one differentiated variety of a product p. So when we refer to product p in the following we do not refer to the firm specific variety, but to a product classification that is produced by many firms in slightly differentiated ways.

We extend the baseline Melitz model only to allow products to differ in complexity and subsequently in the necessary fixed costs to produce/export while not varying any other parameters between single firms or product industries. Apart from product complexity C_p, the baseline environment is kept symmetric across product industries. The derivations below are therefore general Melitz-style cutoff and selection results that apply to any product industry p. Within this product classification firms still produce heterogeneous versions of the same good.

Let c denote the exporting country and p the product industry. The following parameters are common across product industries. Firms are heterogeneous in productivity $\varphi_i > 0$, drawn from a distribution $G(\varphi)$. Demand within each product industry is CES with elasticity of substitution $\sigma > 1$, and firms compete under monopolistic competition. Firms charge a constant markup $\frac{\sigma}{\sigma-1}$ and the wage is normalised to one, so all fixed costs are measured in labour units. Product complexity is denoted by C_p. Exporter-country frictions are denoted by F_c which are common across all product industries within a country. These frictions capture features of the

domestic production and business environment, such as weak contract enforcement, bureaucratic burdens, poor infrastructure, unreliable logistics, limited access to electricity, or administrative uncertainty.

5.1 Fixed costs, complexity, and frictions

Our key primitive assumptions concern fixed domestic operating costs and fixed export costs. Based on the two propositions we have stated in the section above, allow both to vary across product industries and exporter countries:

$$f_{cp}^{d} = f^{d}(C_p, F_c) > 0, \qquad f_{cp}^{x} = f^{x}(C_p, F_c) > 0. \tag{1}$$

For domestic and export fixed costs, we assume

$$\frac{\partial f_c^{d,x}(C_p, F_c)}{\partial C_p} > 0, \qquad \frac{\partial f_c^{d,x}(C_p, F_c)}{\partial F_c} > 0, \qquad \frac{\partial^2 \ln f_c^{d,x}(C_p, F_c)}{\partial C_p \partial F_c} > 0. \tag{2}$$

Thus, complex products require higher fixed domestic operating costs and higher fixed export costs. Exporter-country frictions also raise both types of fixed costs, and the proportional increase is larger for complex products. The log-supermodularity assumption is the central mechanism of the model.[14]

5.2 Domestic production cutoff

Domestic operating profit is

$$\pi_{icp}^{d}(\varphi_i) = \frac{r_{icp}^{d}(\varphi_i)}{\sigma} - f_{cp}^{d}. \tag{3}$$

where domestic revenue is $r_{icp}^{d}(\varphi_i) = \Omega_{cp}^{d}\varphi_i^{\sigma-1}$, with Ω_{cp}^{d} capturing domestic market size, demand, and the domestic price index. A firm produces domestically if $\pi_{icp}^{d}(\varphi_i) \geq 0$. From which follows the domestic productivity cutoff for production:

$$\varphi_{cp}^{d} = \left(\frac{\sigma f_{cp}^{d}}{\Omega_{cp}^{d}}\right)^{\frac{1}{\sigma-1}} \quad \text{or in log terms:} \quad \ln \varphi_{cp}^{d} = \frac{1}{\sigma-1}\left[\ln \sigma + \ln f_{cp}^{d} - \ln \Omega_{cp}^{d}\right]. \tag{4}$$

To see the effect of a rise in frictions we need to take into account that Ω_{cp}^{d} is endogenous. It depends on the price index, which in turn depends on the mass of varieties and aggregate productivity which is affected by fixed costs. Appendix B.1 derives this relationship, and Appendix B.3 shows that, under the Pareto-Melitz benchmark used below, the endogenous response of Ω_{cp}^{d} does not overturn the effect of fixed costs on the cutoff. Under this condition,

$$\frac{\partial^2 \ln \varphi_{cp}^{d}}{\partial C_p \partial F_c} = \frac{1}{\sigma-1}\left[\frac{\partial^2 \ln f_{cp}^{d}}{\partial C_p \partial F_c} - \frac{\partial^2 \ln \Omega_{cp}^{d}}{\partial C_p \partial F_c}\right] > 0. \tag{5}$$

Hence, frictions raise the productivity threshold for domestic production especially strongly for complex products. Appendix B.3 shows that this cost-dominance condition is consistent with the Pareto-Melitz benchmark: the domestic revenue shifter may rise after an increase in fixed costs, but it rises less than proportionally, so $f_{cp}^{d}/\Omega_{cp}^{d}$ increases and the cutoff rises.

5.3 Export cutoff

As in Melitz, exporting requires an additional fixed export cost. Export operating profit is

$$\pi_{icp}^{x}(\varphi_i) = \frac{r_{icp}^{x}(\varphi_i)}{\sigma} - f_{cp}^{x}. \tag{6}$$

[14] Iceberg trade costs may also be affected by exporter-country frictions. Conceptually, however, if iceberg costs are affected by frictions, we assume that this effect is not differentially stronger for complex products. We therefore allow $\partial \ln \tau_{cp}/\partial F_c \geq 0$, but impose $\partial^2 \ln \tau_{cp}/\partial C_p \partial F_c = 0$. We have analysed if applied tariffs (measured by WTO) are correlated with PCI but have seen no correlation that would motivate an argument on the relationship between the iceberg trade costs and complexity.

where export revenue is $r^x_{icp}(\varphi_i) = \Omega^m_{cp}\tau^{1-\sigma}_{cp}\varphi^{\sigma-1}_i$, with Ω^m_{cp} capturing foreign-market demand and destination-side market size, while τ_{cp} and f^x_{cp} capture the exporter-side costs of serving that market. Adopting the small open economy assumption we assume these foreign market conditions to be unaffected by home country frictions and subsequent export changes.[15] A firm exports if $\pi^x_{icp}(\varphi_i) \geq 0$, from which follows the export productivity cutoff:

$$\varphi^x_{cp} = \tau_{cp}\left(\frac{\sigma f^x_{cp}}{\Omega^m_{cp}}\right)^{\frac{1}{\sigma-1}} \quad \text{or in log terms:} \quad \ln\varphi^x_{cp} = \ln\tau_{cp} + \frac{1}{\sigma-1}\left[\ln\sigma + \ln f^x_{cp} - \ln\Omega^m_{cp}\right]. \tag{7}$$

Since Ω^m_{cp} is independent of exporter-country frictions, and since iceberg costs are not assumed to generate a complexity-friction interaction it follows that,

$$\frac{\partial^2 \ln\varphi^x_{cp}}{\partial C_p \partial F_c} = \frac{1}{\sigma-1}\frac{\partial^2 \ln f^x(C_p, F_c)}{\partial C_p \partial F_c} > 0. \tag{8}$$

Thus, frictions raise the export productivity cutoff especially strongly for complex products.

5.4 Exporter Mass

We maintain the standard Melitz exporter-selection case in which the export cutoff exceeds the domestic production cutoff, $\varphi^x_{cp} \geq \varphi^d_{cp}$. Exporters are therefore a subset of active domestic producers. Let M_{cp} denote the mass of active domestic producers in country c and product industry p. Let p^x_{cp} denote the share of these producers that export. The mass of exporters is

$$M^x_{cp} = p^x_{cp}M_{cp} \quad \overset{\text{under Pareto}}{=} \quad \left(\frac{\varphi^d_{cp}}{\varphi^x_{cp}}\right)^{\kappa} M_{cp}. \tag{9}$$

Appendix B.4 derives the exporter share from the conditional productivity distribution, using the Pareto productivity benchmark stated in Appendix B.3. Under this benchmark, where κ is the Pareto shape parameter and $\kappa > \sigma - 1$, equation (9) gives the exporter mass relevant for export presence in a product industry.

Appendix B.4 further shows that, under the Pareto assumption, $M_{cp}(\varphi^d_{cp})^{\kappa}$ is constant with respect to the domestic fixed operating cost. The fall in the mass of active producers and the rise in the domestic productivity cutoff offset each other. Thus,

$$M^x_{cp} \propto (\varphi^x_{cp})^{-\kappa}. \tag{10}$$

Since the export cutoff is given by (7), exporter-country frictions reduce the exporter mass whenever they raise fixed export costs or iceberg trade costs. For the complexity-friction interaction, iceberg costs are not assumed to generate a cross-effect and Ω^m_{cp} is independent of exporter-country frictions. Therefore,

$$\frac{\partial^2 \ln M^x_{cp}}{\partial C_p \partial F_c} = -\kappa\frac{\partial^2 \ln\varphi^x_{cp}}{\partial C_p \partial F_c} = -\frac{\kappa}{\sigma-1}\frac{\partial^2 \ln f^x(C_p, F_c)}{\partial C_p \partial F_c} < 0. \tag{11}$$

Thus, frictions reduce the mass of exporters especially strongly for complex products.

5.4.1 Product Activation

Observed product activation is not a primitive object in Melitz. It is an empirical measurement layer that maps the latent exporter mass into observed country-product export status. We assume that product p is observed as exported by country c if the exporter mass exceeds a small reporting or observation threshold $\mu > 0$:

$$A_{cp} = 1\{M^x_{cp} > \mu\}. \tag{12}$$

[15] The small open economy assumption means domestic changes do not affect world markets. This implies: $\frac{\partial \ln\Omega^m_{cp}}{\partial F_c} = 0$, and $\frac{\partial^2 \ln\Omega^m_{cp}}{\partial C_p \partial F_c} = 0$.

Since product activation is increasing in M^x_{cp}, and since frictions reduce M^x_{cp} especially strongly for complex products, exporter-country frictions reduce product activation especially strongly for complex products (see equation (11)).

Hypothesis 1: Product activation. Exporter-country frictions reduce the likelihood that a country exports a product, and this negative effect is stronger for more complex products.

5.4.2 Product-level export value

Product-level export value is the aggregate export revenue generated:

$$\mathcal{X}_{cp} = M_{cp} \int_{\varphi^x_{cp}}^{\infty} r^x_{icp}(\varphi)\, d\mu_{cp}(\varphi), \tag{13}$$

where $\mu_{cp}(\varphi)$ is the productivity distribution among active domestic producers. Appendix B.5 derives the closed-form expression. Under the Pareto assumption and using the fact that $M_{cp}(\varphi^d_{cp})^\kappa$ is constant with respect to the domestic fixed operating cost, product-level export value can be written as

$$\mathcal{X}_{cp} \propto \Omega^m_{cp} \tau^{1-\sigma}_{cp} (\varphi^x_{cp})^{\sigma-1-\kappa}. \tag{14}$$

Because $\kappa > \sigma - 1$, we have $\sigma - 1 - \kappa < 0$. Therefore, a higher export cutoff reduces product-level export value. Since $1 - \sigma < 0$, higher iceberg trade costs also reduce product-level export value.

Using (7), the small open economy assumption, and the assumption that iceberg costs do not generate a complexity-friction cross-effect, the complexity-friction effect on export value is

$$\frac{\partial^2 \ln \mathcal{X}_{cp}}{\partial C_p \partial F_c} = (\sigma - 1 - \kappa) \frac{\partial^2 \ln \varphi^x_{cp}}{\partial C_p \partial F_c} = \frac{\sigma - 1 - \kappa}{\sigma - 1} \frac{\partial^2 \ln f^x(C_p, F_c)}{\partial C_p \partial F_c} < 0. \tag{15}$$

Thus, exporter-country frictions reduce product-level export value especially strongly for complex products.

Hypothesis 2: Product-level export performance. Exporter-country frictions reduce product-level export value, and this negative effect is stronger for more complex products.

5.5 Interpretation: productivity, welfare, and missing complex products

The model distinguishes between measured productivity and welfare. Complex products require higher fixed operating costs. When such products are produced, only firms with sufficiently high productivity can cover these fixed costs. Thus, complex product industries have higher productivity cutoffs and higher measured productivity among surviving firms.

Higher fixed costs also reduce the mass of active varieties. In Melitz, welfare is real income, $(W = P^{-1})$. Since labour is the only factor of production and the wage is normalised to one, this can also be interpreted as real output per worker and is therefore the model's closest analogue to an aggregate TFP or labour-productivity measure. Appendix B.6 shows that, under the Pareto-Melitz benchmark, higher fixed domestic costs raise measured productivity among surviving firms but lower welfare because the loss of varieties dominates the selection effect.

This distinction matters empirically. Under our assumptions, low-friction countries can sustain more high-fixed-cost complex products, while high-friction countries face fixed costs that rise especially strongly in complex products. Because fixed costs are log-supermodular in complexity and frictions, the model implies lower welfare due to lower product activation and lower export values in complex products for high-friction countries, even if the remaining active firms are more productive on average.

These results imply that institutional frictions generate welfare losses by reducing variety within a product industry and by forcing the highest-productivity, complex product industries to shut down entirely, thereby destroying canonical gains from variety. Remembering the arguments from section 3.4 this supports the idea that the macroeconomic benefits of economic complexity accrue primarily through learning-by-doing and cross-industry knowledge spillovers. The destruction of these frontier sectors can induce aggregate productivity losses.

These negative effects are amplified if we expand our scope to include the role of cronyism in firm-level dynamics, as in the extension to the Melitz model proposed by Bai et al. (2024). If costs are subsidised by political cronyism, firm survival decouples from innate technical efficiency. An increase in fixed costs can trigger perverse selection: politically connected but technologically inferior firms survive, while highly productive, unconnected firms are forced to exit. Aggregate productivity can therefore decline unambiguously, driven by the macroeconomic loss of complex-sector spillovers and the microeconomic reallocation of market share to inefficient incumbents.

6 Empirical Strategy

6.1 Data

Product outcomes are based on UN COMTRADE country-product-year export data. This dataset follows the four-digit HS92 classification and was retrieved from the Harvard Growth Lab (HGL) portal (United Nations Statistics Division, 2026; The Growth Lab at Harvard University, 2026a). Product complexity is measured using the Harvard Growth Lab Product Complexity Index (PCI) (Harvard Growth Lab, 2026). We restrict the analysis to manufacturing products. This restriction is motivated by the fact that manufacturing is the part of the economy most closely aligned with the tradable-sector mechanism studied in our theoretical model. Appendix Figure 8 illustrates the distribution of PCI across different products. Product distance is also obtained from the HGL, and measures the relatedness between a new product and an economy's existing export capabilities (Harvard Growth Lab, 2026).

The final active friction set contains 17 country-level variables from three sources. Institutional frictions come from the World Governance Indicators and include rule of law, control of corruption, regulatory quality, government effectiveness, political stability, and voice and accountability (World Bank, 2026c). Logistical frictions come from the World Bank Logistics Performance Index and include overall logistics, customs, international shipping, infrastructure, logistics quality, tracking, and timeliness (World Bank, 2026a).[16] We further include two infrastructure friction measures from the World Development Indicators. These measure access to electricity and electric power transmission and distribution losses (World Bank, 2026b). All friction variables are coded so that larger values indicate worse institutional, logistical, or infrastructure conditions. We provide a table summarising the coverage in Appendix Table 13.

We construct two principal component (PC) variables, one for the WGI institutional measures and one for the LPI logistics measures. We do this to provide estimates that summarise these two types of frictions. The institutional PC explains 86.6% of the variance in the WGI series and the logistics PC 92.5% of the variance in the LPI series, with roughly equal weighting and uniform signs for all variables inside the PCs. Both indices are oriented so that higher values indicate higher frictions. Appendix figures 6 and 7 show the distribution of both PCs across countries.

For our horse-race robustness specification, we include standard controls for factor scarcity and intensity.[17] Country factor scarcity measures come from Penn World Table 11 and the sector factor intensities come from the WIOD Socio Economic Accounts and are assigned to HS4 products through a product-sector bridge (Groningen

[16] Because the LPI is observed only in survey waves, the active LPI series are linearly interpolated between survey years and endpoint-carried within the constructed LPI panel.

[17] We reorient factor abundance so that higher values denote skill or capital scarcity.

Growth and Development Centre, 2025; World Input–Output Database, 2021).[18] We acknowledge that this product-sector mapping is approximate and should be interpreted as a limitation of the horse-race controls. For the alternative FE-specifications we further obtain human capital, capital per worker, real output-side GDP, real output-side GDP per capita, the investment share of GDP, and domestic credit to the private sector as a share of GDP from the Penn World Table 11 as controls.

6.2 Regressions

Our baseline specification is

$$Y_{cpt} = \beta \left(F_{c,t-1} \times PCI_{p,t-3:t-5} \right) + \delta Distance_{cp,t-1} + \alpha_{cp} + \gamma_{ct} + \eta_{pt} + \varepsilon_{cpt}. \tag{16}$$

The outcome Y_{cpt} is one of three product-level export outcomes: log exports, positive exporter status, and an indicator for whether the product accounts for more than 0.001% of total country-year exports.[19] The activation regressions are estimated using a linear probability model. $F_{c,t-1}$ represents a country-level friction lagged by one year. The product complexity term is a moving average of product PCI at lags three, four, and five. Lagged product distance is included only in the export activation specifications. This ensures that the estimated friction-complexity interaction is not simply picking up the fact that complex products are often farther from countries' existing capabilities.

The coefficient β measures whether a one-standard-deviation increase in friction is associated with a differential change in the outcome of a product that is one standard deviation more complex. A negative and significant β would provide evidence supporting our assumption of log-supermodularity in fixed costs, implying that higher-friction countries perform relatively worse in more complex products than in less complex products. Following our hypothesis, we expect a negative β.

Country-product fixed effects, α_{cp}, absorb time-invariant determinants of a country's exports in a given product. Country-year fixed effects, γ_{ct}, absorb all country-wide shocks and country-level variables in a given year. Product-year fixed effects, η_{pt}, absorb global product shocks, changes in world demand, and the average effect of product complexity in each year. Thus, the coefficient of interest is identified from the changes in country-level frictions interacted with complexity, net of country-year, product-year, and country-product variation. For all estimates, standard errors are clustered two ways by country and product, allowing for arbitrary correlation within countries over products and time, and within products across countries and time.

The timing of the variables is intended to reduce issues with simultaneity. The friction measure is lagged by one year, making it predetermined relative to current export outcomes. Product complexity is measured using PCI from years $t-3$ to $t-5$, reducing the concern that contemporaneous export performance mechanically affects the complexity measure used in the regression. These choices do not eliminate endogeneity concerns, since many frictions are persistent and evolve with broader country development. For that reason our estimates are interpreted as conditional associations instead of pure causal effects.

We also estimate the following horse-race specification:

$$\begin{aligned} Y_{cpt} = \beta \left(F_{c,t-1} \times PCI_{p,t-3:t-5} \right) + \theta_s \left(S_{c,t-1} \times s_p \right) + \theta_k \left(K_{c,t-1} \times k_p \right) \\ + \delta Distance_{cp,t-1} + \alpha_{cp} + \gamma_{ct} + \eta_{pt} + \varepsilon_{cpt}. \end{aligned} \tag{17}$$

Here, $S_{c,t-1}$ denotes skill scarcity and $K_{c,t-1}$ capital scarcity at the country-year level, while s_p and k_p denote product-level skill and capital intensity. The purpose of this specification is not to replace the baseline identi-

[18] See Appendix Table 14

[19] This threshold serves as a robustness check as it eliminates noise from small and inconsequential export volumes that are recorded in export data for completeness. We verify that the threshold does not distort or truncate the distribution of the PCI compared to the original distribution without thresholds.

fication strategy, but to assess whether the friction-complexity coefficient remains stable after controlling for traditional factor-endowment forces. Under a standard comparative-advantage mechanism, countries that are relatively scarce in skill or capital should perform worse in products that use those factors intensively. If the coefficient on $F_{c,t-1} \times PCI_{p,t-3:t-5}$ remains negative and similar in magnitude after including these interactions, this suggests that the complexity result is not simply a relabeling of skill- or capital-endowment comparative advantage.

7 Results

This section reports the main product-level evidence. We first present the baseline estimates, then assess robustness to factor-endowment controls and alternative fixed-effect structures.

Table 2: Headline product-level estimates

	Outcome variable			Sample	
Country-level friction	Log exports	Positive exporter	Exporter share $> 0.001\%$	Obs.	Coverage countries; years
Institutional Frictions	-0.109^{***} [0.025]	-0.006^{***} [0.002]	-0.004^{**} [0.002]	4,466,366	204 1999–2024
Logistical Frictions	-0.057^{***} [0.021]	-0.004^{**} [0.002]	-0.002^{*} [0.001]	2,824,454	170 2008–2024

Notes: Cells report coefficients from product-level regressions of export outcomes on standardized country-level frictions interacted with standardized lagged product complexity. Two-way clustered standard errors are reported in square brackets. All specifications include country–product, country–year, and product–year fixed effects. The dependent variables are log HGL exports, an indicator for positive exports, and an indicator for export shares above 0.001 percent. The coverage column reports the number of countries and regression sample window. Observation counts and coverage vary across frictions because source coverage differs. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 2 reports the baseline product-level estimates of the friction–complexity interaction. Across the main country-level friction measures, the interaction coefficient is consistently negative, in line with the prediction that frictions negatively impact complex products more strongly. For log exports, a one-standard-deviation increase in institutional frictions is associated with an additional decline of 0.109 log points for products that are one standard deviation more complex. The corresponding estimate for logistical frictions is 0.057 log points. Interpreted in percentage terms, these estimates imply relative export losses of approximately 10.3% and 5.6%. In Table 18, for log exports, the institutional-friction interaction coefficient of -0.109 is about 11% of the estimated direct institutional-friction penalty of -1.032 log points, suggesting that exposure to high product complexity applies a unique, non-trivial penalty to complex sectors on top of the independent effect of institutional frictions. We find weak evidence for a significant main effect of logistical frictions on log exports, consistent with the idea that exporting firms manage to internalise the costs of managing their own logistics.

The activation outcomes show the same pattern: higher institutional and logistical frictions are associated with lower probabilities of exporting more complex products. In the baseline specification, the coefficient on institutional frictions interaction is -0.006 for positive exporter status, while the coefficient on logistical frictions interaction is -0.004. This means that a one-standard-deviation increase of frictions reduces by 0.6 and 0.4 percentage points, respectively, the probability of exporting a good that is one standard deviation more complex. Although these magnitudes appear small in absolute terms, they are not negligible relative to the scale of the main effects recovered in the fixed-effect decomposition. In Table 18, the positive-exporter main effect of institutional frictions is -0.079. In that sense, the institutional frictions interaction coefficient of -0.006 is close to the estimated complexity main effect and equal to an additional penalty of 8% over and above the independent effect of institutional frictions. For logistical frictions, the interaction estimate of -0.004 adds an extra penalty of 44% over the independent effect of logistical frictions of -0.009.

Taken together, these results support both hypotheses derived from the model. First, the negative coefficients for positive exporter status indicate that country-level frictions reduce product activation more strongly for complex products. Second, the negative coefficients for log exports indicate that frictions also reduce product-

level export performance more strongly in complex products. The evidence is therefore consistent with the model's central mechanism: fixed costs appear to be effectively log-supermodular in product complexity and country-level frictions, so that frictions act as an increasing tax on complexity rather than as a uniform penalty across all products.

Table 3 shows that the main results are robust to alternative specifications. Adding the horse-race controls for skill scarcity interacted with skill intensity and capital scarcity interacted with capital intensity does not weaken the estimates. The coefficients are also stable when product-year fixed effects, country-year fixed effects, or both sets of time-varying fixed effects are removed. The cross-sectional estimates are substantially larger, but these should be interpreted more cautiously because they rely more heavily on between-country and between-product variation. The key result is that the negative interaction remains present in the saturated panel specification and survives the inclusion of factor-endowment controls. This supports the interpretation that the estimates are not simply capturing standard comparative advantage in skill- or capital-intensive products, but rather a differential sensitivity of complex products to institutional and logistical frictions.

Table 3: Robustness to Alternative Specifications

	Base-line (1)	+ HR controls (2)	No prod. -year FE (3)	No ctry. -year FE (4)	No time FE (5)	Cross-section (6)
Panel A. Log exports						
Institutional Frictions	-0.109^{***}	-0.137^{***}	-0.137^{***}	-0.136^{***}	-0.141^{***}	-0.518^{***}
	[0.025]	[0.027]	[0.026]	[0.028]	[0.028]	[0.044]
Logistical Frictions	-0.057^{***}	-0.063^{***}	-0.053^{**}	-0.057^{**}	-0.049^{**}	-0.559^{***}
	[0.021]	[0.022]	[0.022]	[0.024]	[0.024]	[0.040]
Panel B. Positive exporter						
Institutional Frictions	-0.006^{***}	-0.008^{***}	-0.008^{***}	-0.008^{***}	-0.009^{***}	-0.024^{***}
	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]	[0.003]
Logistical Frictions	-0.004^{**}	-0.004^{**}	-0.004^{**}	-0.004^{**}	-0.004^{**}	-0.026^{***}
	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]	[0.004]
HR controls	No	Yes	Yes	Yes	Yes	Yes
Product-year FE	Yes	Yes	No	Yes	No	No
Country-year FE	Yes	Yes	Yes	No	No	No
Panel data	Yes	Yes	Yes	Yes	Yes	No
Cross-section	No	No	No	No	No	Yes

Notes: Cells report the β coefficient on the interaction $\text{Friction}_{t-1} \times \text{PCI}_{t-3:t-5}$. Standard errors are in square brackets. All regressions are at the product level. HR controls are skill scarcity interacted with skill intensity and capital scarcity interacted with capital intensity. Dropped-FE specifications additionally control for human capital, capital per worker, real output-side GDP, real output-side GDP per capita, investment/GDP, and domestic credit/GDP. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

8 Discussion and Conclusion

This paper began with a familiar problem in the growth literature: much of the variation in income between countries is contained within the productivity residual. Our aim was to open this "black box" by studying whether economic complexity can help explain cross-country variation in TFP. The empirical evidence presented in our motivating facts supports this interpretation. Countries with higher ECI scores have higher levels of TFP on average, and initial complexity predicts subsequent ten-year TFP growth. These relationships survive standard macroeconomic, geographic, and institutional controls. The variance decomposition points in the same direction: complexity accounts for a meaningful share of cross-country income variation, and an even larger share of cross-country variation in TFP. A central implication of our findings is that institutions and complexity should not be treated as competing explanations of development. Institutional quality attenuates the ECI coefficient, but does not eliminate its significance, suggesting that each variable captures distinct but related dimensions of the same process.

The remainder of the paper studied this link between institutions and complexity at the product level. Our theoretical framework treats fixed costs of producing and exporting as increasing functions of frictions and complexity. We find that institutional and logistical frictions are relatively more damaging to complex products than simple ones. These results are robust to controlling for standard factor-endowment comparative advantage mechanisms. These findings are in line with our argument that complex products require more complementary tasks, more relationship-specific inputs, more coordination, and more reliable execution, rather than being constrained by factor scarcity alone.

Frictions act as an increasing tax on complexity. Their cost is not only that they reduce average economic performance. They also distort the composition of production away from activities that require more coordination, contracting, logistics, finance, and up-front investment. In that sense, we cannot state that complexity alone causes development, nor that institutions alone explain complexity. Rather, development appears to require the joint accumulation of productive capabilities and the institutional environment that allows those capabilities to be used. Our results do not resolve the chain of causality. Instead, they support the narrower claim that institutions and productive capabilities may be complementary. Economic complexity is useful because it reveals this productive structure; frictions matter because they shape which parts of that structure can emerge.

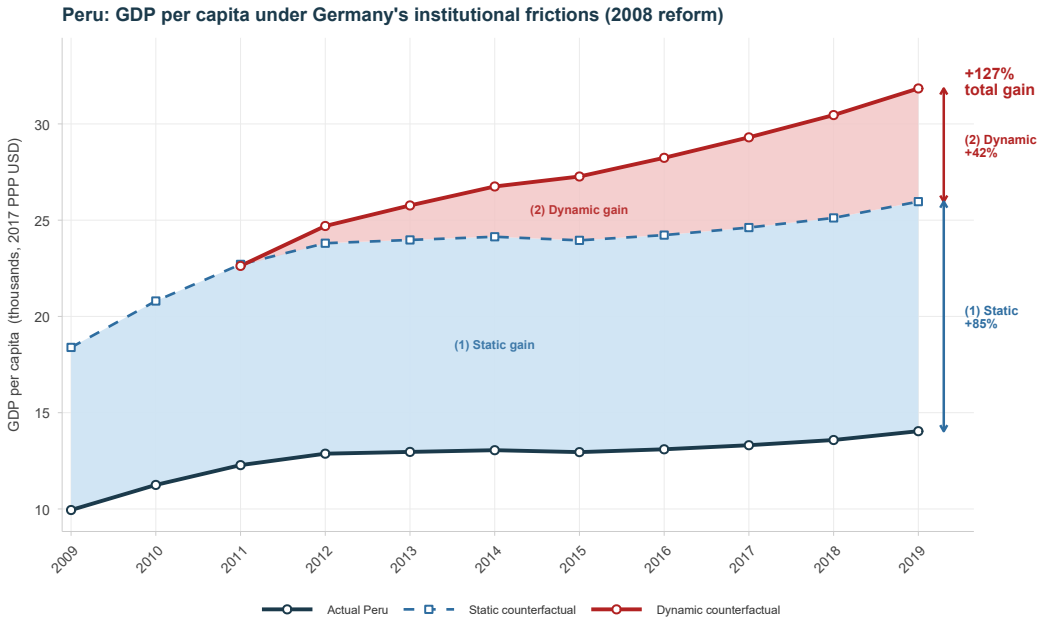


Figure 5: Counterfactual growth trajectory of Peru with Germany's institutions starting from 2009.

To conclude, we return to the observation from Lucas (1988), that differences in income between countries open our eyes to widely different possibilities for growth. Based on our analysis, we compute a rough counterfactual exercise to make the implications of our findings tangible.[20] If a developing country like Peru successfully lowered its frictions to the level of an advanced economy like Germany, what would that mean for Peru's growth possibilities? Better institutional and logistical quality would immediately lead to a static increase in the level of TFP, but they also enable a more complex basket of exports, which in turn would lead to sustained and dynamic TFP growth. Precluding any increase in physical or human capital, Peru's income per capita would more than double, leaving it in the company of such celebrated case studies of growth as South Korea and Japan in just ten years, quantifying just how much is at stake when discussing frictions, economic complexity, and productivity growth.

Echoing the sentiment of this paper's opening quote, we are compelled to exercise a similar level of restraint in interpreting our findings. While we have presented robust evidence that productive capabilities and economic

[20] Details on the specific assumptions made for this back-of-the-envelope exercise are provided in the Appendix G.

complexity carry meaningful explanatory power for cross-country divergence in income and total factor productivity, we do not claim to have offered a panacea for the continued challenges surrounding growth and development. The temptation in empirical work of this kind is to overreach; to mistake statistical findings for structural understanding and policy-ready conclusions. What we offer instead is a modest contribution to an ongoing conversation; a contribution that narrows, by some small measure, the considerable gap between what we observe and what we can credibly claim to understand about why some countries are richer than others.

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A Data, Methods and Robustness (Empirical Patterns)

This appendix documents the data, variable construction, and methodology underlying the motivating facts of Section 3. Appendix A.1 describes sources. Appendix A.2 details the construction of total factor productivity, including the perpetual-inventory capital stock and the choice of capital share. Appendix A.3 describes the construction of our institutional control. Appendix A.4 sets out the variance-decomposition methodology behind Figure 3 and reports the full set of decomposition results. Appendix A.5 reports the regression robustness checks, including the extreme-bounds analysis, capital-share sensitivity, regional heterogeneity.

A.1 Data sources and sample

Our estimation sample is an unbalanced country–year panel covering 124 countries. Coverage begins in 1970, when the SITC-based ECI series first attains broad country coverage, and ends in 2021, the last year for which all controls are jointly available; the natural-resource series from the World Development Indicators carries a roughly two-year publication lag, which binds the right edge of the panel. To hold the sample fixed across specifications, we restrict to the set of countries present in both the full levels regression and the full growth regression. Table 4 lists each variable, its precise definition, and its source.

Table 4: Variable definitions and sources

Variable	Definition	Source
Outcomes and factors		
$\ln y$	Log real output per worker, $\ln(\text{rgdpo}/\text{emp})$	PWT 11.0
$\ln A$	Log TFP residual from development accounting, Eq. (19)	PWT 11.0; own calc.
$\ln k$	Log capital per worker, $\ln(K^{\text{PIM}}/\text{emp})$	PWT 11.0; own calc.
$\ln h$	Log of the PWT human-capital index (`hc`)	PWT 11.0
$\Delta \ln A$	Ten-year change in $\ln A$	Own calc.
Variable of interest		
ECI	Economic Complexity Index, SITC vintage, standardised	Atlas of Econ. Complexity
Controls		
NR rents	Natural-resource rents as a share of GDP (NY.GDP.TOTL.RT.ZS/100)	WDI
Investment/GDP	Investment share of output (`csh_i`)	PWT 11.0
Exports/GDP	Export share of output (`csh_x`)	PWT 11.0
Government/GDP	Government-consumption share of output (`csh_g`)	PWT 11.0
Log population	$\ln(\text{population})$	WDI
Inflation	Annual CPI inflation, % (FP.CPI.TOTL.ZG)	WDI
Institutions	First principal component of six WGI series (Appendix A.3)	WGI
Absolute latitude	Absolute latitude of the capital, $\lvert\text{lat}\rvert$	CEPII GeoDist
Landlocked	Indicator for landlocked country	CEPII GeoDist

The ECI is computed from different trade classifications. We use the SITC-based index as our headline measure because it provides the longest panel (from 1970). To confirm that the index choice is immaterial, Table 5 compares the SITC series with the HS12 and HS92 variants over their overlapping windows. They are nearly indistinguishable in levels (pooled correlations above 0.97) and strongly correlated within countries over time (median within-country correlations above 0.88), so our results are not an artefact of the classification used. We additionally include Life Expectancy from the WDI as an additional control cited by (Sala-i Martin, 1997) as an important control in a cross-country setting as a proxy for levels of healthcare and confirm ECI robustness, however its correlation with Institutional Quality precludes its inclusion in the headline results.

Table 5: ECI vintage equivalence

Pair	Overlap window	N obs	Pooled r	Median within-r
SITC vs. HS12	2012–2023	1,692	0.978	0.889
SITC vs. HS92	1995–2023	4,058	0.974	0.904
HS12 vs. HS92	2012–2023	1,692	1.000	0.990

Note: within-country correlations require at least five overlapping years; computed on 141 countries.

A.2 Development accounting and the construction of TFP

We construct TFP as the Solow residual implied by a Cobb–Douglas technology with constant returns to scale. Output per worker is

$$y_{ct} = A_{ct}\, k_{ct}^{\alpha}\, h_{ct}^{1-\alpha}, \tag{18}$$

where k_{ct} is physical capital per worker, h_{ct} the human-capital index, and α the capital share. Taking logs gives the log TFP residual,

$$\ln A_{ct} = \ln y_{ct} - \alpha \ln k_{ct} - (1-\alpha) \ln h_{ct}. \tag{19}$$

We set $\alpha = 1/3$ following Hall and Jones (1999) and Caselli (2005). Output per worker uses output-side real GDP (`rgdpo`) divided by employment, and the human-capital index is the PWT measure `hc`.

A.2.1 Perpetual-inventory capital stock.

Physical capital is built by the perpetual-inventory method (PIM). With real investment $I_{ct} = \text{rgdpo}_{ct} \cdot \text{csh_i}_{ct}$, the stock evolves as

$$K_{ct} = (1-\delta) K_{c,t-1} + I_{ct}, \qquad \delta = 0.06, \tag{20}$$

with the depreciation rate following **?**. The initial stock is seeded from the steady-state relationship

$$K_{c,t_0} = \frac{I_{c,t_0}}{g_c + \delta}, \tag{21}$$

where the trend investment growth rate g_c is estimated from a log-linear regression of investment on time over each country's first ten years of positive investment. Following Caselli (2005), the first ten years of each country's constructed stock are treated as unreliable due to being dominated by the initialisation assumption rather than by observed investment, and are excluded from all estimation. This filter retains roughly 84% of country–year observations. As a check, we verify our results against an alternative Solow-transition PIM that backs out capital from the model's transition path; the two capital series deliver materially identical TFP estimates.

A.2.2 Capital-share robustness.

Equation (19) is linear in α, so the residual at any alternative share α' can be recomputed directly:

$$\ln A_{ct}(\alpha') = \ln y_{ct} - \alpha' \ln k_{ct} - (1-\alpha') \ln h_{ct}. \tag{22}$$

The ten-year growth-accounting outcome is constructed analogously,

$$\Delta \ln A_{ct} = \Delta \ln y_{ct} - \alpha\, \Delta \ln k_{ct} - (1-\alpha)\, \Delta \ln h_{ct}, \qquad \Delta \ln x_{ct} \equiv \ln x_{ct} - \ln x_{c,t-10}. \tag{23}$$

Sensitivity of the ECI coefficient to $\alpha \in \{0.30, 1/3, 0.40\}$ and to country–year shares $\alpha_{ct} = 1 - \text{labsh}_{ct}$ is reported in Appendix A.5.

A.3 The Institutional Control

We summarise institutional quality with the first principal component of the six Worldwide Governance Indicators (World Bank, 2026c): voice and accountability, political stability and absence of violence, government effectiveness, regulatory quality, rule of law, and control of corruption. Each series is standardised before extraction. The first component explains 86.6% of the common variance and loads positively and almost equally on all six indicators (Table 6), consistent with a single latent "institutional quality" dimension. We orient the component so that higher values denote better institutions, anchoring on the sign of the rule-of-law loading.

Table 6: Principal-component structure of the WGI

WGI series	PC1 loading
Voice and accountability	0.392
Political stability & absence of violence	0.370
Government effectiveness	0.418
Regulatory quality	0.419
Rule of law	0.428
Control of corruption	0.420
Variance explained (PC1)	86.6%

Because the WGI begin in 1996, specifications that include institutions are estimated on a shorter panel. This accounts for the fall in sample size between the macro and full specifications in Table 1 (for the growth panel, from 4,095 to 1,777 observations).

A.4 Variance decomposition

To quantify economic importance (the third motivating fact and Figure 3) we follow the development-accounting decomposition of Caselli (2005). In a given cross-section, the log of output per worker is written as the sum of a factor term and the fitted contributions of additional components. Each component j contributes $\hat{c}_j = \hat{\beta}_j x_j$, where the $\hat{\beta}_j$ come from a regression of the factor-adjusted residual on the components. We summarise explanatory power with three standard statistics.

The variance-ratio ("Success-1") statistic is

$$S_j^{(1)} = \frac{\mathrm{Var}(\hat{c}_j)}{\mathrm{Var}(\ln y)}; \tag{24}$$

the dispersion (90–10) statistic is

$$S_j^{(2)} = \frac{\exp\big(Q_{0.9}(\hat{c}_j) - Q_{0.1}(\hat{c}_j)\big)}{\exp\big(Q_{0.9}(\ln y) - Q_{0.1}(\ln y)\big)}; \tag{25}$$

and the covariance share, following Shorrocks (1982) as applied by Klenow and Rodríguez-Clare (1997), is

$$S_j^{(\mathrm{KR})} = \frac{\mathrm{Cov}(\hat{c}_j, \ln y)}{\mathrm{Var}(\ln y)}. \tag{26}$$

The covariance share is our preferred measure because, unlike $S^{(1)}$, it is additive across components and invariant to the order in which correlated regressors enter, a material advantage given the high correlation between ECI and institutions ($r = 0.75$). Figure 3 reports $S^{(\mathrm{KR})}$, and the bands sum to unity by construction.

Table 7 reports all three statistics for our focal cross-section, for Caselli's decomposition (target $\ln y$). This includes with and without institutions. The ranges in square brackets are the minimum and maximum across all cross-sections in the common sample.

Table 7: Variance decomposition of cross-country income, all metrics

Specification	Component	$S^{(1)}$	$S^{(2)}$	$S^{(\mathrm{KR})}$
Caselli static, factors + ECI				
	Factors (K, H)	0.301 [0.275, 0.329]	0.285 [0.234, 0.343]	0.522 [0.475, 0.552]
	ECI	0.083 [0.044, 0.121]	0.125 [0.093, 0.175]	0.199 [0.126, 0.267]
Caselli static, factors + ECI + institutions				
	Factors (K, H)	0.301 [0.275, 0.329]	0.285 [0.234, 0.343]	0.522 [0.475, 0.552]
	ECI	0.013 [0.000, 0.031]	0.075 [0.045, 0.113]	0.071 [−0.001, 0.133]
	Institutions	0.059 [0.036, 0.113]	0.114 [0.084, 0.140]	0.179 [0.141, 0.258]

Note: point estimates are for the focal cross-section (2016); square brackets give [min, max] across all years in the common 2002–2021 sample. Both specifications are estimated on a common sample so that the with- and without-institutions results are directly comparable.

Two patterns are worth noting. First, the covariance share of ECI in the income decomposition is sizeable (up to 27%) but falls to an upper bound of about 13% once institutions enter, reflecting the variance the two share. Second, the three metrics tell a consistent story: across $S^{(1)}$, $S^{(2)}$, and $S^{(\mathrm{KR})}$, ECI's contribution is non-trivial but smaller than that of factor accumulation, and is attenuated by institutions in each case. While both factors possess distinct, statistically significant marginal returns (as shown by our regression results), their intense historical co-accumulation means that cross-sectional development accounting exercises assign their explanatory power almost entirely to their joint covariance structure, leading to a lower bound of 0 share explained by ECI in some years once institutions are introduced.

A.5 Regression robustness

A.5.1 Complete Regression Results:

Table 8: Economic Complexity and Total Factor Productivity

	TFP levels: $\log A_{i,t}$			Ten-year TFP growth: $\Delta \log A_{i,t}$		
	Base	+ Macro & others	Full model	Convergence	+ Macro & others	Full model
	(1)	(2)	(3)	(4)	(5)	(6)
ECI	0.384***	0.330***	0.184**	0.129***	0.128***	0.102***
	(0.035)	(0.034)	(0.056)	(0.020)	(0.025)	(0.032)
Initial TFP, $\log A_{i,t-10}$				−0.291***	−0.330***	−0.407***
				(0.035)	(0.048)	(0.086)
Natural resource rents	2.021***	1.594***	2.026***	0.297**	0.346**	0.459*
	(0.354)	(0.308)	(0.329)	(0.142)	(0.141)	(0.235)
Investment/GDP		−0.337	−0.366		−0.063	0.004
		(0.275)	(0.335)		(0.178)	(0.197)
Inflation		−0.00004	−0.0001·		0.00001	0.00007*
		(0.00003)	(0.00006)		(0.00002)	(0.00004)
Log population		−0.072**	−0.048*		−0.016*	−0.011
		(0.022)	(0.022)		(0.009)	(0.012)
Exports/GDP		0.226**	0.076		0.092*	0.056
		(0.075)	(0.069)		(0.047)	(0.059)
Government/GDP		0.058	−0.224		−0.013	0.071
		(0.333)	(0.422)		(0.176)	(0.248)
Landlocked		−0.346***	−0.322***		−0.071*	−0.057
		(0.065)	(0.066)		(0.038)	(0.043)
Institutional quality			0.080***			0.015
			(0.019)			(0.022)
Absolute latitude			0.005*			
			(0.002)			
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Clustered SEs	Country	Country	Country	Country	Country	Country
Observations	5,472	5,063	2,745	4,491	4,095	1,777
R^2	0.400	0.518	0.606	0.248	0.267	0.262
Within R^2	0.384	0.505	0.597	0.166	0.173	0.183

Notes: Columns (1)–(3) report regressions of log total factor productivity, $\log A_{i,t}$, on contemporaneous ECI and controls. Columns (4)–(6) report regressions of ten-year TFP growth, $\Delta \log A_{i,t}$, on initial ECI and initial controls. ECI is the SITC-based Economic Complexity Index. All specifications include year fixed effects. Standard errors, clustered at the country level, are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$, $^{\cdot}p < 0.10$.

A.5.2 Extreme-bounds analysis.

We implement a Levine–Renelt style extreme-bounds exercise to assess whether the ECI coefficient's sign and significance is sensitive to a broad range of cross-country controls from the literature:

$$\log A_{i,t} = \alpha + \beta \, \mathrm{ECI}_{i,t} + \mathbf{I}'_{i,t}\boldsymbol{\gamma} + \mathbf{Z}'_{i,t}\boldsymbol{\eta} + \delta_t + \varepsilon_{i,t}. \tag{3}$$

Here, $\mathbf{I}_{i,t}$ is the vector of always-included controls and $\mathbf{Z}_{i,t}$ is a subset of controls drawn from the rotating pool. Following Levine and Renelt (1992), we estimate the equation over all subsets of the rotating pool with up to three controls, yielding a distribution of $\hat{\beta}$ estimates. The Levine–Renelt criterion classifies the ECI coefficient as robust if $\hat{\beta}$ retains its sign and remains statistically significant at the 5 percent level across every specification.

Table 9: Extreme-bounds analysis: control sets

Specification	Always-included controls (I)	Rotating pool (Z, choose ≤ 3)
Levels ($\ln A$)	NR rents, investment/GDP, log population, inflation	exports/GDP, government/GDP, absolute latitude, landlocked, institutions
Growth ($\Delta \ln A$)	initial $\ln A$, initial NR rents, initial investment/GDP, initial log population, initial inflation	initial institutions, initial exports/GDP, initial government/GDP, landlocked

Note: all specifications include year fixed effects; standard errors are clustered by country.

Table 10: Extreme-bounds analysis: results

	Lower bound	Upper bound	Mean $\hat{\beta}$	Median $\hat{\beta}$	Robust?
TFP levels ($\ln A$)	0.081	0.478	0.284	0.291	Yes
TFP growth ($\Delta \ln A$)	0.030	0.204	0.111	0.112	Yes

Note: bounds are the minimum lower and maximum upper values of $\hat{\beta} \pm 2\,\mathrm{SE}$ across all control subsets. The ECI coefficient is strictly positive in every specification, satisfying the Levine–Renelt criterion.

A.5.3 Capital-share sensitivity and capital structure.

Table 11 re-estimates the full growth specification under alternative capital shares as well as a capital-intensity dummy (an indicator for countries with above-median mean capital share, $1 - \text{labsh}$). The ECI coefficient is essentially invariant across capital shares, moving only from 0.081 at $\alpha = 0.30$ to 0.080 at $\alpha = 0.40$, and is unaffected by controlling for differences in capital intensity.

Table 11: Capital-share sensitivity, ten-year TFP growth

	$\alpha = 0.30$	$\alpha = 1/3$	$\alpha = 0.40$	$\alpha = 1/3$, +cap. int.
Initial ECI	0.081**	0.081**	0.080**	0.080**
	(0.033)	(0.033)	(0.032)	(0.033)
N	1,522	1,522	1,522	1,522

Note: full control set, year fixed effects, standard errors clustered by country. The final column adds an above-median capital-intensity indicator (coefficient not significant). $^{**}p < 0.05$.

A.5.4 Regional heterogeneity.

Table 12 adds World Bank regional dummies (relative to East Asia & Pacific) to the growth specification. The ECI coefficient remains positive and significant through the base and macro specifications (0.085) and attenuates only modestly, to 0.060, in the full specification with institutions and geography.

Table 12: Growth specification with regional dummies

	(1) Base	(2) +Macro &other	(3) Full
Initial ECI	0.085***	0.085***	0.060*
	(0.022)	(0.026)	(0.036)
Regional dummies	Yes	Yes	Yes
N	4,614	4,170	1,777

Note: dummies relative to East Asia & Pacific; year fixed effects; standard errors clustered by country. $^{*}p < 0.10$, $^{***}p < 0.01$.

B Adapted Melitz-Model derivations

B.1 CES demand, pricing, and revenue shifters

This appendix collects the model derivations used in the main text. The derivations are for a product-indexed Melitz industry p in country c. Within each product industry, consumers have CES preferences over a continuum of differentiated varieties. Let $\mathcal{V}_{cp}$ denote the set of available varieties in product industry p. Product-level quantity is

$$Q_{cp} = \left[\int_{\omega\in\mathcal{V}_{cp}} q(\omega)^{\rho} d\omega\right]^{1/\rho}, \qquad 0<\rho<1, \qquad \sigma = \frac{1}{1-\rho} > 1. \tag{27}$$

The corresponding price index is

$$P_{cp} = \left[\int_{\omega\in\mathcal{V}_{cp}} p(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}. \tag{28}$$

Consumer demand implies revenue

$$r_{cp}(\omega) = R_{cp}\left(\frac{p(\omega)}{P_{cp}}\right)^{1-\sigma}, \tag{29}$$

where R_{cp} is product-level expenditure.

Production uses labour only. A firm with productivity φ_i has marginal cost $1/\varphi_i$. Under monopolistic competition with CES demand, the firm charges a constant markup over marginal cost:

$$p(\varphi_i) = \frac{1}{\rho\varphi_i}, \qquad \rho = \frac{\sigma-1}{\sigma}. \tag{30}$$

Substituting (30) into (29) gives domestic revenue

$$r^d_{icp}(\varphi_i) = R_{cp}(P_{cp}\rho\varphi_i)^{\sigma-1} = \Omega^d_{cp}\varphi_i^{\sigma-1}, \tag{31}$$

where

$$\Omega^d_{cp} = R_{cp}(P_{cp}\rho)^{\sigma-1}. \tag{32}$$

Thus Ω^d_{cp} is not a primitive technology parameter. It is a market-side revenue shifter that depends on domestic expenditure and the domestic price index.

For exports, iceberg trade costs imply that $\tau_{cp} > 1$ units must be shipped for one unit to arrive abroad. With foreign-market demand summarised by Ω^m_{cp}, export revenue is

$$r^x_{icp}(\varphi_i) = \Omega^m_{cp}\tau_{cp}^{1-\sigma}\varphi_i^{\sigma-1}. \tag{33}$$

The term $\tau_{cp}^{1-\sigma}$ is decreasing in iceberg trade costs because $1-\sigma<0$.

B.2 Domestic and export cutoff derivations

Domestic operating profit is

$$\pi^d_{icp}(\varphi_i) = \frac{r^d_{icp}(\varphi_i)}{\sigma} - f^d_{cp}. \tag{34}$$

A firm produces domestically if $\pi^d_{icp}(\varphi_i) \geq 0$. The marginal domestic producer satisfies

$$\frac{\Omega^d_{cp}(\varphi^d_{cp})^{\sigma-1}}{\sigma} = f^d_{cp}. \tag{35}$$

Solving gives

$$\varphi_{cp}^{d} = \left(\frac{\sigma f_{cp}^{d}}{\Omega_{cp}^{d}}\right)^{\frac{1}{\sigma-1}}, \qquad \ln \varphi_{cp}^{d} = \frac{1}{\sigma-1}\left[\ln \sigma + \ln f_{cp}^{d} - \ln \Omega_{cp}^{d}\right]. \tag{36}$$

Taking the cross-partial with respect to complexity and frictions gives

$$\frac{\partial^2 \ln \varphi_{cp}^{d}}{\partial C_p \partial F_c} = \frac{1}{\sigma-1}\left[\frac{\partial^2 \ln f_{cp}^{d}}{\partial C_p \partial F_c} - \frac{\partial^2 \ln \Omega_{cp}^{d}}{\partial C_p \partial F_c}\right]. \tag{37}$$

Therefore, frictions raise the domestic production cutoff especially strongly for complex products if the fixed-cost interaction dominates the induced response of the domestic revenue shifter:

$$\frac{\partial^2 \ln f_{cp}^{d}}{\partial C_p \partial F_c} > \frac{\partial^2 \ln \Omega_{cp}^{d}}{\partial C_p \partial F_c}. \tag{38}$$

Appendix B.3 shows that this condition is satisfied in the Pareto-Melitz benchmark used in the main text.

Export operating profit is

$$\pi_{icp}^{x}(\varphi_i) = \frac{r_{icp}^{x}(\varphi_i)}{\sigma} - f_{cp}^{x}. \tag{39}$$

The marginal exporter satisfies

$$\frac{\Omega_{cp}^{m} \tau_{cp}^{1-\sigma} (\varphi_{cp}^{x})^{\sigma-1}}{\sigma} = f_{cp}^{x}. \tag{40}$$

Solving gives

$$\varphi_{cp}^{x} = \tau_{cp}\left(\frac{\sigma f_{cp}^{x}}{\Omega_{cp}^{m}}\right)^{\frac{1}{\sigma-1}}, \qquad \ln \varphi_{cp}^{x} = \ln \tau_{cp} + \frac{1}{\sigma-1}\left[\ln \sigma + \ln f_{cp}^{x} - \ln \Omega_{cp}^{m}\right]. \tag{41}$$

Since Ω_{cp}^{m} is independent of exporter-country frictions and iceberg costs are not assumed to generate a complexity-friction interaction,

$$\frac{\partial^2 \ln \varphi_{cp}^{x}}{\partial C_p \partial F_c} = \frac{1}{\sigma-1}\frac{\partial^2 \ln f^{x}(C_p, F_c)}{\partial C_p \partial F_c} > 0. \tag{42}$$

B.3 Price index, aggregate productivity, and the Pareto-Melitz benchmark

Let M_{cp} denote the mass of domestic varieties available in product industry p, and let $\mu_{cp}(\varphi)$ denote the productivity distribution among active firms. The Melitz aggregate productivity index is

$$\widetilde{\varphi}_{cp} = \left[\int_{\varphi_{cp}^{d}}^{\infty} \varphi^{\sigma-1} d\mu_{cp}(\varphi)\right]^{\frac{1}{\sigma-1}}. \tag{43}$$

Using the pricing rule, the product-level inverse price index can be written as

$$P_{cp}^{-1} = \rho M_{cp}^{\frac{1}{\sigma-1}} \widetilde{\varphi}_{cp}. \tag{44}$$

Thus product-level real income, or product-level welfare, is

$$W_{cp} = P_{cp}^{-1} = \rho M_{cp}^{\frac{1}{\sigma-1}} \widetilde{\varphi}_{cp}. \tag{45}$$

Taking logs gives

$$d \ln W_{cp} = \frac{1}{\sigma-1} d \ln M_{cp} + d \ln \widetilde{\varphi}_{cp}. \tag{46}$$

This expression makes clear that welfare depends both on the mass of available varieties and on the measured productivity of surviving firms.

Using (44) in (32), the domestic revenue shifter can be rewritten as

$$\Omega_{cp}^{d} = R_{cp} M_{cp}^{-1} \widetilde{\varphi}_{cp}^{1-\sigma}. \tag{47}$$

Hence

$$d \ln \Omega_{cp}^{d} = d \ln R_{cp} - d \ln M_{cp} - (\sigma - 1) d \ln \widetilde{\varphi}_{cp}. \tag{48}$$

A fall in the mass of varieties tends to raise the revenue available to surviving varieties, while a rise in aggregate productivity tends to lower the revenue shifter through the price index.

For the closed-form results, assume that productivity is Pareto distributed:

$$1 - G(\varphi) = \left(\frac{\varphi_{\min}}{\varphi} \right)^{\kappa}, \qquad \kappa > \sigma - 1. \tag{49}$$

Here $\varphi_{\min}$ is the lower-bound productivity parameter. The condition $\kappa > \sigma - 1$ ensures that the relevant productivity moment is finite.

Under Pareto, the Melitz aggregate productivity index is proportional to the domestic production cutoff:

$$\widetilde{\varphi}_{cp} = B \varphi_{cp}^{d}, \qquad B = \left(\frac{\kappa}{\kappa - (\sigma - 1)} \right)^{\frac{1}{\sigma - 1}}. \tag{50}$$

The ratio $\widetilde{\varphi}_{cp} / \varphi_{cp}^{d}$ is therefore constant under Pareto.

We now show explicitly why the endogenous response of the domestic revenue shifter does not overturn the fixed-cost channel in the Pareto-Melitz benchmark. Consider an increase in the fixed domestic operating cost f_{cp}^{d}, holding product expenditure R_{cp}, the sunk entry cost, and the productivity distribution fixed. The zero-cutoff-profit condition implies that average operating profit is proportional to the fixed operating cost:

$$\bar{\pi}_{cp} = f_{cp}^{d} \left[\left(\frac{\widetilde{\varphi}_{cp}}{\varphi_{cp}^{d}} \right)^{\sigma - 1} - 1 \right]. \tag{51}$$

Since $\widetilde{\varphi}_{cp} / \varphi_{cp}^{d}$ is constant under Pareto, the term in brackets is constant and therefore $\bar{\pi}_{cp} \propto f_{cp}^{d}$.

The Melitz free-entry condition implies that expected operating profits cover the sunk entry cost. Since only a fraction $1 - G(\varphi_{cp}^{d})$ of entrants survive the domestic cutoff, the free-entry condition can be written as

$$\bar{\pi}_{cp} = \frac{\delta f^{e}}{1 - G(\varphi_{cp}^{d})}, \tag{52}$$

where f^{e} is the sunk entry cost and δ is the exogenous exit probability. Under the Pareto distribution in (49),

$$\bar{\pi}_{cp} = \delta f^{e} \left(\frac{\varphi_{cp}^{d}}{\varphi_{\min}} \right)^{\kappa}. \tag{53}$$

Combining (51) and (53) gives

$$(\varphi_{cp}^{d})^{\kappa} \propto f_{cp}^{d}. \tag{54}$$

Therefore,

$$\frac{d \ln \varphi_{cp}^{d}}{d \ln f_{cp}^{d}} = \frac{1}{\kappa}. \tag{55}$$

Since aggregate productivity is proportional to the cutoff,

$$\frac{d \ln \widetilde{\varphi}_{cp}}{d \ln f_{cp}^{d}} = \frac{1}{\kappa} > 0. \tag{56}$$

Average revenue is

$$\bar{r}_{cp} = \sigma(\bar{\pi}_{cp} + f_{cp}^{d}), \tag{57}$$

so $\bar{r}_{cp} \propto f_{cp}^{d}$. Because aggregate product expenditure satisfies

$$R_{cp} = M_{cp} \bar{r}_{cp}, \tag{58}$$

holding R_{cp} fixed implies

$$M_{cp} \propto \frac{1}{f_{cp}^{d}}, \qquad \frac{d \ln M_{cp}}{d \ln f_{cp}^{d}} = -1. \tag{59}$$

This elasticity follows from the Pareto-Melitz benchmark with fixed product expenditure, fixed sunk entry costs, and a fixed productivity distribution. It is not an unconditional result of every Melitz-type environment.

Using (36) and (55), the response of the domestic revenue shifter is

$$\frac{d \ln \Omega_{cp}^{d}}{d \ln f_{cp}^{d}} = 1 - \frac{\sigma - 1}{\kappa}. \tag{60}$$

Because $\kappa > \sigma - 1$,

$$0 < \frac{d \ln \Omega_{cp}^{d}}{d \ln f_{cp}^{d}} < 1. \tag{61}$$

Hence, the domestic revenue shifter may rise when fixed costs increase, but it rises less than proportionally. Therefore $f_{cp}^{d} / \Omega_{cp}^{d}$ increases and the domestic production cutoff rises.

This also delivers the domestic cutoff interaction result used in the main text. From (60), the Pareto-Melitz benchmark implies

$$\ln \Omega_{cp}^{d} = \text{constant} + \left(1 - \frac{\sigma - 1}{\kappa}\right) \ln f_{cp}^{d}. \tag{62}$$

Therefore,

$$\frac{\partial^2 \ln \Omega_{cp}^{d}}{\partial C_p \partial F_c} = \left(1 - \frac{\sigma - 1}{\kappa}\right) \frac{\partial^2 \ln f_{cp}^{d}}{\partial C_p \partial F_c}. \tag{63}$$

Substituting (63) into the general cutoff interaction in (37) gives

$$\frac{\partial^2 \ln \varphi_{cp}^{d}}{\partial C_p \partial F_c} = \frac{1}{\kappa} \frac{\partial^2 \ln f_{cp}^{d}}{\partial C_p \partial F_c} > 0. \tag{64}$$

Thus, under the Pareto-Melitz benchmark, the endogenous response of the domestic revenue shifter does not overturn the fixed-cost channel. Instead, the log-supermodularity of domestic fixed costs in complexity and frictions is transmitted into a positive complexity-friction interaction in the domestic production cutoff.

B.4 Exporter mass derivation

We maintain the standard Melitz exporter-selection case in which the export cutoff exceeds the domestic production cutoff, $\varphi_{cp}^{x} \geq \varphi_{cp}^{d}$. This is satisfied if

$$\tau_{cp} \left(\frac{f_{cp}^{x}}{\Omega_{cp}^{m}} \right)^{\frac{1}{\sigma - 1}} \geq \left(\frac{f_{cp}^{d}}{\Omega_{cp}^{d}} \right)^{\frac{1}{\sigma - 1}}, \tag{65}$$

where the common σ term cancels from both sides. Under this condition, exporters are a subset of active domestic producers.

Since only firms with productivity above φ_{cp}^x export, and active producers are firms with productivity above φ_{cp}^d, the exporter share is

$$p_{cp}^x = \Pr(\varphi_i \geq \varphi_{cp}^x \mid \varphi_i \geq \varphi_{cp}^d) = \frac{1 - G(\varphi_{cp}^x)}{1 - G(\varphi_{cp}^d)}. \tag{66}$$

Using the Pareto distribution in (49), this becomes

$$p_{cp}^x = \left(\frac{\varphi_{cp}^d}{\varphi_{cp}^x}\right)^{\kappa}. \tag{67}$$

Therefore,

$$M_{cp}^x = M_{cp}\left(\frac{\varphi_{cp}^d}{\varphi_{cp}^x}\right)^{\kappa}. \tag{68}$$

The Pareto-Melitz benchmark gives

$$M_{cp} \propto \frac{1}{f_{cp}^d}, \qquad \varphi_{cp}^d \propto (f_{cp}^d)^{1/\kappa}. \tag{69}$$

Hence,

$$M_{cp}(\varphi_{cp}^d)^{\kappa} = \text{constant with respect to } f_{cp}^d. \tag{70}$$

The fall in the mass of active producers and the rise in the domestic productivity cutoff offset each other in the exporter-mass expression. Thus,

$$M_{cp}^x \propto (\varphi_{cp}^x)^{-\kappa}. \tag{71}$$

Taking the cross-partial gives

$$\frac{\partial^2 \ln M_{cp}^x}{\partial C_p \partial F_c} = -\kappa \frac{\partial^2 \ln \varphi_{cp}^x}{\partial C_p \partial F_c} = -\frac{\kappa}{\sigma - 1} \frac{\partial^2 \ln f^x(C_p, F_c)}{\partial C_p \partial F_c} < 0. \tag{72}$$

B.5 Product-level export value derivation

Product-level export value is aggregate export revenue generated by exporting firms:

$$\mathcal{X}_{cp} = M_{cp} \int_{\varphi_{cp}^x}^{\infty} r_{icp}^x(\varphi)\, d\mu_{cp}(\varphi), \tag{73}$$

where $\mu_{cp}(\varphi)$ is the productivity distribution among active domestic producers. Since active producers are those with $\varphi \geq \varphi_{cp}^d$,

$$d\mu_{cp}(\varphi) = \frac{dG(\varphi)}{1 - G(\varphi_{cp}^d)} \quad \text{for} \quad \varphi \geq \varphi_{cp}^d. \tag{74}$$

Substituting export revenue gives

$$\mathcal{X}_{cp} = M_{cp}\Omega_{cp}^m \tau_{cp}^{1-\sigma} \int_{\varphi_{cp}^x}^{\infty} \varphi^{\sigma-1}\, d\mu_{cp}(\varphi). \tag{75}$$

Under Pareto,

$$\int_{\varphi_{cp}^x}^{\infty} \varphi^{\sigma-1}\, d\mu_{cp}(\varphi) = \frac{\kappa}{\kappa - (\sigma - 1)} (\varphi_{cp}^d)^{\kappa} (\varphi_{cp}^x)^{\sigma-1-\kappa}. \tag{76}$$

Therefore,

$$\mathcal{X}_{cp} \propto M_{cp}\Omega_{cp}^m \tau_{cp}^{1-\sigma} (\varphi_{cp}^d)^{\kappa} (\varphi_{cp}^x)^{\sigma-1-\kappa}. \tag{77}$$

Using (70), product-level export value can be written as

$$\mathcal{X}_{cp} \propto \Omega_{cp}^{m} \tau_{cp}^{1-\sigma} (\varphi_{cp}^{x})^{\sigma-1-\kappa}. \tag{78}$$

Because $\kappa > \sigma - 1$, the exponent $\sigma - 1 - \kappa$ is negative. Hence, a higher export cutoff reduces product-level export value.

Using (41), the small open economy assumption, and the assumption that iceberg costs do not generate a complexity-friction cross-effect,

$$\frac{\partial^2 \ln \mathcal{X}_{cp}}{\partial C_p \partial F_c} = (\sigma - 1 - \kappa) \frac{\partial^2 \ln \varphi_{cp}^{x}}{\partial C_p \partial F_c} = \frac{\sigma - 1 - \kappa}{\sigma - 1} \frac{\partial^2 \ln f^{x}(C_p, F_c)}{\partial C_p \partial F_c} < 0. \tag{79}$$

B.6 Fixed costs, measured productivity, and welfare

The Pareto benchmark clarifies the relationship between fixed costs and measured productivity. Because

$$\widetilde{\varphi}_{cp} = B \varphi_{cp}^{d}, \tag{80}$$

and

$$\frac{d \ln \varphi_{cp}^{d}}{d \ln f_{cp}^{d}} = \frac{1}{\kappa}, \tag{81}$$

we have

$$\frac{d \ln \widetilde{\varphi}_{cp}}{d \ln f_{cp}^{d}} = \frac{1}{\kappa} > 0. \tag{82}$$

Thus, a product industry with higher fixed domestic operating costs has higher measured productivity among surviving firms. This does not mean that higher fixed costs improve firm technology. Rather, higher fixed costs raise the productivity cutoff, so only more productive firms can profitably operate.

Since product complexity raises fixed domestic operating costs, the same logic implies that complex products have higher measured productivity when they are produced:

$$C_p \uparrow \Rightarrow f_{cp}^{d} \uparrow \Rightarrow \varphi_{cp}^{d} \uparrow \Rightarrow \widetilde{\varphi}_{cp} \uparrow . \tag{83}$$

This is a selection effect. Complex products are associated with higher measured productivity because only more productive firms can cover their higher fixed costs. It does not require complex products to embody superior firm-level technology.

Combining the productivity response in (56) with the mass response in (59) gives

$$\frac{d \ln W_{cp}}{d \ln f_{cp}^{d}} = \frac{1}{\kappa} - \frac{1}{\sigma - 1} < 0, \tag{84}$$

because $\kappa > \sigma - 1$. Thus, higher fixed domestic operating costs raise measured productivity among surviving firms, but reduce product-level welfare because the loss of varieties dominates the selection-driven productivity gain.

To describe aggregate producer productivity across product industries, define a product-weighted productivity index

$$\ln \widetilde{\Phi}_c = \sum_p \omega_{cp} \ln \widetilde{\varphi}_{cp}, \qquad \sum_p \omega_{cp} = 1, \tag{85}$$

where ω_{cp} are production or expenditure weights. If higher-complexity products have higher $\widetilde{\varphi}_{cp}$, then an economy with greater weight on complex product industries has higher measured aggregate producer

productivity:

$$\sum_p \omega_{cp} C_p \quad \text{higher} \quad \Rightarrow \quad \ln \widetilde{\Phi}_c \quad \text{higher}, \tag{86}$$

provided that the mapping from complexity to fixed costs, and from fixed costs to cutoffs, is positive as described above. This aggregate productivity comparison is again a selection and composition result.

To aggregate welfare across product industries, assume an upper-level CES aggregator:

$$U_c = \left[\sum_p \alpha_p^{1/\chi} Q_{cp}^{\frac{\chi-1}{\chi}} \right]^{\frac{\chi}{\chi-1}}, \qquad \chi > 0, \qquad \chi \neq 1, \qquad \sum_p \alpha_p = 1, \tag{87}$$

where χ is the elasticity of substitution across product industries and α_p are taste weights. We use an upper-level CES aggregator because it is the standard generalisation of Cobb-Douglas: it allows product industries to be imperfectly substitutable while nesting the Cobb-Douglas case as the limit $\chi \to 1$.

Within each product industry, preferences remain CES over differentiated varieties as in the product-level Melitz environment. The upper-level CES aggregator therefore does not change the within-industry demand system, pricing rule, or cutoff derivations. It only determines how expenditure is allocated across product industries. The corresponding aggregate price index is

$$P_c = \left[\sum_p \alpha_p P_{cp}^{1-\chi} \right]^{\frac{1}{1-\chi}}, \qquad \chi \neq 1, \tag{88}$$

and the expenditure share of product industry p is

$$s_{cp} = \alpha_p \left(\frac{P_{cp}}{P_c} \right)^{1-\chi}. \tag{89}$$

Thus, relative to Cobb-Douglas, product expenditure shares are allowed to adjust when relative product-level price indices change. This affects the product-level revenue shifter Ω_{cp}^d through expenditure R_{cp}, and we summarise this general-equilibrium demand response inside Ω_{cp}^d rather than solving for it explicitly.[21] The product-level cutoff results are therefore interpreted conditional on the relevant revenue shifters, together with the cost-dominance condition that the induced response of Ω_{cp}^d does not overturn the fixed-cost channel.

With nominal income normalised, aggregate real income is

$$W_c = P_c^{-1}. \tag{90}$$

For local changes, the CES price index implies

$$d \ln W_c = \sum_p s_{cp} d \ln W_{cp}. \tag{91}$$

Using the product-level welfare expression,

$$d \ln W_c = \sum_p s_{cp} \left[\frac{1}{\sigma - 1} d \ln M_{cp} + d \ln \widetilde{\varphi}_{cp} \right]. \tag{92}$$

[21] To see this, recall that the domestic revenue shifter can be written as $\Omega_{cp}^d = R_{cp}(P_{cp}\rho)^{\sigma-1}$. Under the upper-level CES aggregator, product expenditure is $R_{cp} = s_{cp} E_c$, with $s_{cp} = \alpha_p (P_{cp}/P_c)^{1-\chi}$. Holding nominal expenditure E_c fixed,

$$d \ln \Omega_{cp}^d = (1-\chi)(d \ln P_{cp} - d \ln P_c) + (\sigma - 1) d \ln P_{cp}.$$

Thus, the upper-level CES aggregator affects Ω_{cp}^d through the endogenous expenditure share. If product industries are substitutes, $\chi > 1$, then an increase in the relative product-level price index P_{cp}/P_c lowers expenditure on product p. This expenditure-share channel reduces Ω_{cp}^d and therefore reinforces the fixed-cost channel by making f_{cp}^d/Ω_{cp}^d rise more strongly. The cost-dominance condition used above is therefore conservative in this case.

Aggregate welfare falls if

$$\sum_{p} s_{cp} \left[\frac{1}{\sigma - 1} d \ln M_{cp} + d \ln \widetilde{\varphi}_{cp} \right] < 0. \tag{93}$$

Thus, aggregate welfare falls when the expenditure-share-weighted loss of varieties across product industries dominates the expenditure-share-weighted productivity gains among surviving firms.

This expression also clarifies why a low-friction economy may have both higher measured productivity and higher welfare. If low frictions allow a country to sustain high-fixed-cost complex product industries, then the country gains access to additional varieties and to industries with higher measured productivity among surviving firms. However, an increase in frictions has the opposite effect. Because fixed costs are log-supermodular in complexity and frictions, higher frictions disproportionately raise the fixed costs of complex products. As a result, complex products are disproportionately likely to disappear from the produced and exported variety set. This loss reduces welfare through M_{cp}, even though the remaining active firms are more strongly selected and therefore more productive.

C Friction variables: sources and coverage

Table 13: Friction variables: sources and coverage

Friction	Source	Coverage	Construction / coding
Rule of law	World Governance Indicators	206 countries; 24 years (1999–2024); 4,533,550 product-year obs.	Negative of WGI governance estimate; higher values indicate weaker rule of law.
Control of corruption	World Governance Indicators	206 countries; 24 years (1999–2024); 4,491,386 product-year obs.	Negative of WGI governance estimate; higher values indicate weaker control of corruption.
Regulatory quality	World Governance Indicators	204 countries; 24 years (1999–2024); 4,480,314 product-year obs.	Negative of WGI governance estimate; higher values indicate weaker regulatory quality.
Government effectiveness	World Governance Indicators	204 countries; 24 years (1999–2024); 4,478,732 product-year obs.	Negative of WGI governance estimate; higher values indicate weaker government effectiveness.
Political stability	World Governance Indicators	206 countries; 24 years (1999–2024); 4,517,422 product-year obs.	Negative of WGI governance estimate; higher values indicate lower political stability.
Voice and accountability	World Governance Indicators	206 countries; 24 years (1999–2024); 4,524,840 product-year obs.	Negative of WGI governance estimate; higher values indicate weaker voice and accountability.
LPI overall logistics	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI overall score, multiplied by -1; higher values indicate weaker logistics performance.
LPI customs	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI customs score, multiplied by -1; higher values indicate weaker customs performance.
LPI international shipments	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI international shipments score, multiplied by -1.
LPI infrastructure	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI infrastructure score, multiplied by -1.
LPI logistics quality	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI logistics quality score, multiplied by -1.
LPI tracking and tracing	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI tracking and tracing score, multiplied by -1.
LPI timeliness	World Bank Logistics Performance Index	170 countries; 17 years (2008–2024); 2,824,454 product-year obs.	Interpolated annual LPI timeliness score, multiplied by -1.
Electricity access gap	World Development Indicators	208 countries; 27 years (1998–2024); 5,054,961 product-year obs.	Access to electricity as a share of population, multiplied by -1; higher values indicate lower access.
Electric power losses	World Development Indicators	149 countries; 27 years (1998–2024); 3,969,637 product-year obs.	Electric power transmission and distribution losses as a share of output; higher values indicate greater losses.

Notes: Coverage reports the realized product-level regression sample for the active thesis run. All friction variables enter the regressions with a one-year lag and are standardized within each specification. Product-year observations are country–HS4–year cells after applying the relevant outcome, friction, complexity, fixed-effect, and control-variable availability restrictions.

D HS92-ISIC.Rev4 Concordance Table

Table 14: HS92-to-sector bridge for horse-race factor intensities

ISIC Rev. 4	Sector	HS92 chapters assigned	HS4 products	Effective HS4 weight	Mapping rule
10	Food products	02–05, 07–12, 15–21	164	164.0	Direct chapter assignment
11	Beverages	22	9	9.0	Direct chapter assignment
12	Tobacco products	24	3	3.0	Direct chapter assignment
13	Textiles	50–60, 63	115	115.0	Direct chapter assignment
14	Wearing apparel	61–62	34	34.0	Direct chapter assignment
15	Leather and related products	41–43, 64	27	27.0	Direct chapter assignment
16	Wood and products of wood and cork, except furniture	44–46	27	27.0	Direct chapter assignment
17	Paper and paper products	47–48	30	30.0	Direct chapter assignment
18	Printing and reproduction of recorded media	49	11	11.0	Direct chapter assignment
19	Coke and refined petroleum products	27	16	16.0	Direct chapter assignment
20	Chemicals and chemical products	28–29, 31–38	170	170.0	Direct chapter assignment
21	Basic pharmaceutical products and preparations	30	6	6.0	Direct chapter assignment
22	Rubber and plastics products	39–40	43	43.0	Direct chapter assignment
23	Other non-metallic mineral products	25–26, 68–70	100	100.0	Direct chapter assignment
24	Basic metals	71–72, 74–76, 78–81	123	123.0	Direct chapter assignment
25	Fabricated metal products, except machinery and equipment	73, 82–83	52	52.0	Direct chapter assignment
26	Computer, electronic and optical products	85, 90–91	95	71.0	HS 85 weight 0.5
27	Electrical equipment	85	48	24.0	HS 85 weight 0.5
28	Machinery and equipment n.e.c.	84	85	85.0	Direct chapter assignment
29	Motor vehicles, trailers and semi-trailers	87	16	16.0	Direct chapter assignment
30	Other transport equipment	86, 88–89	22	22.0	Direct chapter assignment
31	Furniture	94	6	6.0	Direct chapter assignment
32	Other manufacturing	65–67, 92–93, 95–97	62	62.0	Direct chapter assignment
33	Repair and installation of machinery and equipment	None	0	0.0	–

E PCI and PC-frictions distributions

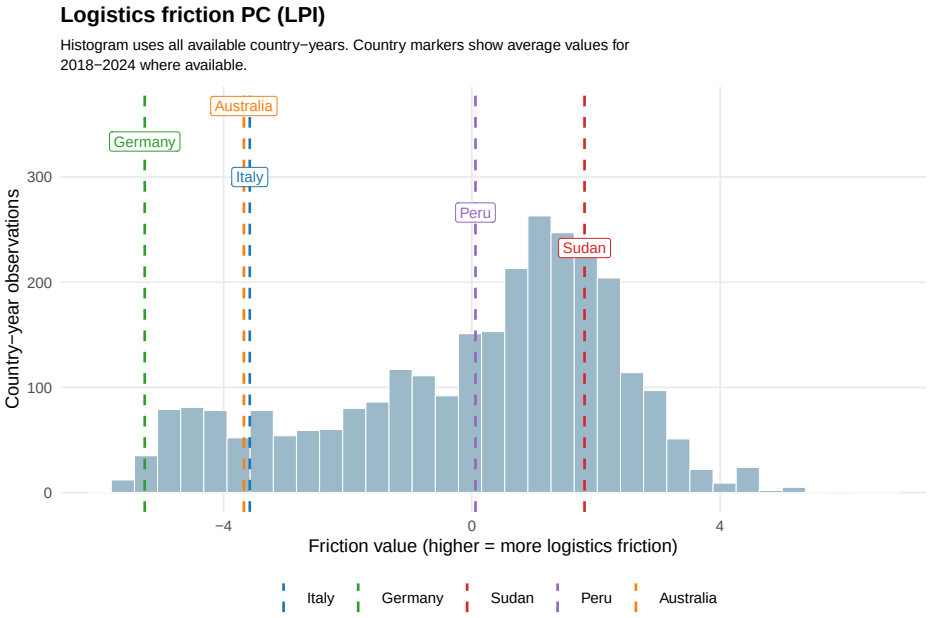


Figure 6: Logistical friction principal component

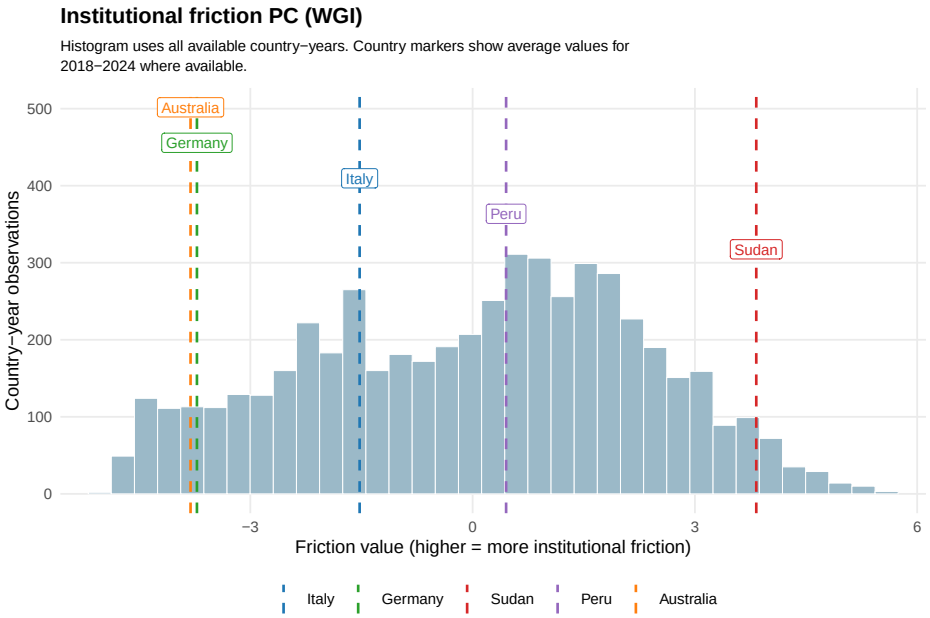


Figure 7: Institutional friction principal component

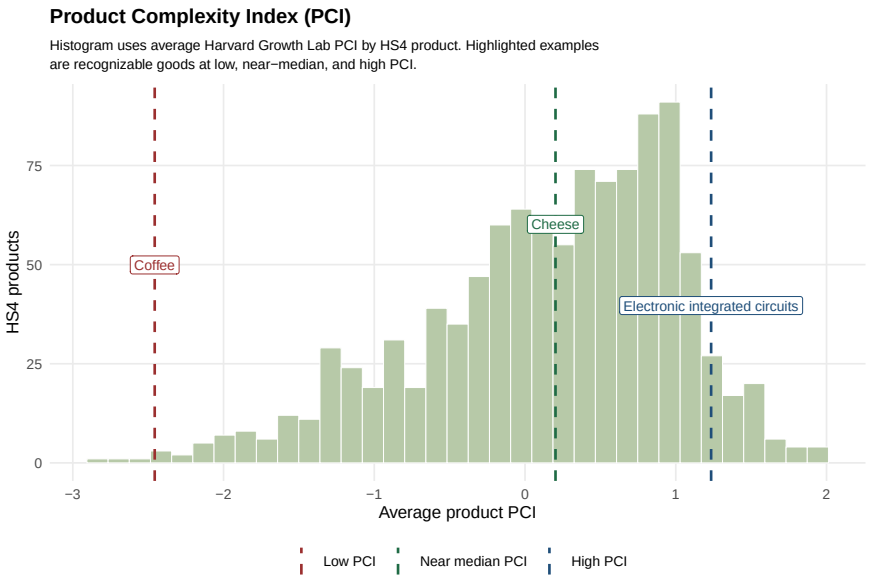


Figure 8: Product complexity across HS92 four-digit products

F Full product-level regressions

Table 15: Full product-level baseline results

Friction	Log exports	Positive exporter	Exporter share $> 0.001\%$	N	Countries	Window
Logistics Performance Index (LPI)						
LPI logistics PC	−0.0573*** [0.0208]	−0.0039** [0.0015]	−0.0025* [0.0014]	2,824,454	170	2008–2024
LPI overall	−0.0582*** [0.0208]	−0.0040** [0.0015]	−0.0025* [0.0014]	2,824,454	170	2008–2024
Customs	−0.0529*** [0.0183]	−0.0036** [0.0015]	−0.0030** [0.0013]	2,824,454	170	2008–2024
Infrastructure	−0.0498*** [0.0178]	−0.0025* [0.0013]	−0.0016 [0.0013]	2,824,454	170	2008–2024
International shipping	−0.0183 [0.0148]	−0.0017 [0.0012]	−0.0013 [0.0010]	2,824,454	170	2008–2024
Logistics quality	−0.0476*** [0.0151]	−0.0029** [0.0011]	−0.0011 [0.0012]	2,824,454	170	2008–2024
Timeliness	−0.0271* [0.0148]	−0.0023** [0.0011]	−0.0010 [0.0010]	2,824,454	170	2008–2024
Tracking and tracing	−0.0481*** [0.0159]	−0.0031*** [0.0012]	−0.0021** [0.0011]	2,824,454	170	2008–2024
World Development Indicators (WDI)						
Electricity access	−0.0485* [0.0264]	−0.0068*** [0.0022]	−0.0056** [0.0022]	5,054,961	208	1998–2024
Power losses	−0.0200 [0.0127]	−0.0019* [0.0010]	−0.0007 [0.0008]	3,969,637	149	1998–2024
Worldwide Governance Indicators (WGI)						
Corruption	−0.1016*** [0.0224]	−0.0049*** [0.0016]	−0.0047*** [0.0016]	4,491,386	206	1999–2024
Government effectiveness	−0.0842*** [0.0208]	−0.0045*** [0.0015]	−0.0032** [0.0016]	4,478,732	204	1999–2024
WGI institutional PC	−0.1086*** [0.0248]	−0.0062*** [0.0019]	−0.0045** [0.0019]	4,466,366	204	1999–2024
Political stability	−0.0370** [0.0165]	−0.0024* [0.0014]	−0.0001 [0.0013]	4,517,422	206	1999–2024
Regulatory quality	−0.0808*** [0.0218]	−0.0050*** [0.0017]	−0.0038** [0.0017]	4,480,314	204	1999–2024
Rule of law	−0.0937*** [0.0212]	−0.0050*** [0.0017]	−0.0038** [0.0016]	4,533,550	206	1999–2024

Continued on next page

Table continued

Friction	Log exports	Positive exporter	Exporter share $> 0.001\%$	N	Countries	Window
Voice/accountability	-0.0921^{***} [0.0189]	-0.0056^{***} [0.0016]	-0.0052^{***} [0.0015]	4,524,840	206	1999–2024

Notes: Cells report product-level interaction coefficients. Standard errors are shown in square brackets. The baseline specifications include the full fixed-effects structure used in the headline table. Frictions are grouped by data family. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 16: Full product-level horse-race results

Friction	Log exports	Positive exporter	Exporter share $> 0.001\%$	N	Countries	Window
Logistics Performance Index (LPI)						
LPI logistics PC	-0.0631^{***} [0.0221]	-0.0040^{**} [0.0016]	-0.0029^{**} [0.0015]	2,397,588	140	2008–2024
LPI overall	-0.0637^{***} [0.0221]	-0.0040^{**} [0.0016]	-0.0029^{**} [0.0015]	2,397,588	140	2008–2024
Customs	-0.0507^{**} [0.0200]	-0.0032^{**} [0.0016]	-0.0027^{*} [0.0014]	2,397,588	140	2008–2024
Infrastructure	-0.0514^{***} [0.0188]	-0.0021 [0.0014]	-0.0020 [0.0013]	2,397,588	140	2008–2024
International shipping	-0.0267^{*} [0.0151]	-0.0021^{*} [0.0012]	-0.0017^{*} [0.0010]	2,397,588	140	2008–2024
Logistics quality	-0.0513^{***} [0.0164]	-0.0027^{**} [0.0012]	-0.0012 [0.0013]	2,397,588	140	2008–2024
Timeliness	-0.0324^{**} [0.0157]	-0.0026^{**} [0.0012]	-0.0015 [0.0010]	2,397,588	140	2008–2024
Tracking and tracing	-0.0555^{***} [0.0179]	-0.0033^{***} [0.0013]	-0.0028^{**} [0.0011]	2,397,588	140	2008–2024
World Development Indicators (WDI)						
Electricity access	-0.0881^{***} [0.0265]	-0.0109^{***} [0.0019]	-0.0091^{***} [0.0022]	3,764,652	143	1998–2024
Power losses	-0.0223 [0.0139]	-0.0021^{**} [0.0010]	-0.0007 [0.0009]	3,498,793	129	1998–2024
Worldwide Governance Indicators (WGI)						
Corruption	-0.1309^{***} [0.0227]	-0.0065^{***} [0.0017]	-0.0062^{***} [0.0018]	3,451,459	144	1999–2024
Government effectiveness	-0.1209^{***} [0.0241]	-0.0065^{***} [0.0018]	-0.0049^{***} [0.0018]	3,451,459	144	1999–2024
WGI institutional PC	-0.1365^{***} [0.0265]	-0.0078^{***} [0.0021]	-0.0059^{***} [0.0020]	3,451,459	144	1999–2024
Political stability	-0.0468^{**} [0.0181]	-0.0034^{**} [0.0016]	-0.0004 [0.0014]	3,451,459	144	1999–2024
Regulatory quality	-0.1153^{***} [0.0239]	-0.0066^{***} [0.0019]	-0.0058^{***} [0.0017]	3,451,459	144	1999–2024
Rule of law	-0.1228^{***} [0.0237]	-0.0067^{***} [0.0020]	-0.0054^{***} [0.0018]	3,451,459	144	1999–2024
Voice/accountability	-0.1037^{***} [0.0217]	-0.0063^{***} [0.0018]	-0.0056^{***} [0.0017]	3,451,459	144	1999–2024

Notes: Cells report product-level interaction coefficients. Standard errors are shown in square brackets. Horse-race specifications include PWT11 controls. Frictions are grouped by data family. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

G Counterfactual Exercise Methodology

The counterfactual depicted in Figure 5 asks: what GDP per capita path would Peru have followed had it adopted Germany's institutional and logistical environment in 2008, with everything else held fixed? This appendix describes the three-step construction: the friction shock, the static path (Scenario A), and the dynamic path (Scenario B). All factor inputs are held at their historically observed levels; the entire counterfactual operates through TFP.

G.1 Step 1: Translating the Friction Gap into a TFP Shock

We use two composite friction indices from Part II: the first principal component of the six World Governance Indicators (z^{inst}, oriented so that higher values indicate better governance) and the first principal component of the five Logistics Performance Index dimensions (z^{LPI}). The *BOTH* bundle uses both simultaneously.

Polarity bridge. In Part II, frictions are coded so that higher values indicate worse conditions. Part I regression (3) uses z^{inst} coded in the opposite direction (higher = better). We therefore set

$$\Delta z^{\text{inst}} = -\Delta f^{\text{inst}},$$

where $\Delta f^{\text{inst}} = f^{\text{inst}}_{\text{PER,2008}} - f^{\text{inst}}_{\text{DEU,2008}}$ is the raw friction gap from the Part II data. The same sign convention applies to Δz^{LPI}.

Cross-calibration. Evaluated at the 2008 base year,

$$z^{\text{inst}}_{\text{DEU}} = 3.789, \qquad z^{\text{inst}}_{\text{PER}} = -0.608, \qquad \Delta z^{\text{inst}} = 4.397.$$

The direct log-TFP effect of closing this gap is

$$\Delta \ln A^{\text{direct}} = \hat{\beta}_{\text{inst}} \cdot \Delta z^{\text{inst}} + \hat{\beta}_{\text{LPI}} \cdot \Delta z^{\text{LPI}}, \tag{94}$$

with $\hat{\beta}_{\text{inst}} = 0.080$ and $\hat{\beta}_{\text{LPI}} =$ [to be filled].

Note on identification. No Part I specification includes both z^{inst} and z^{LPI} simultaneously. Each coefficient is therefore drawn from a *separate* full-model levels regression: $\hat{\beta}_{\text{inst}} = 0.080$ comes from column (3) of Table 1, which controls for the WGI composite alone; $\hat{\beta}_{\text{LPI}}$ comes from an analogous specification in which the LPI composite replaces the WGI composite as the institutional control (reported in Appendix Table **??**). Because the two coefficients are estimated on different samples and specifications, equation (94) should be read as an additive approximation rather than a joint structural estimate.

G.2 Step 2: Static Counterfactual (Scenario A)

Scenario A treats $\Delta \ln A^{\text{direct}}$ as a permanent, time-invariant level shift applied to the actual GDP per capita series y_t from PWT 11.0:

$$\tilde{y}^A_t = y_t \cdot \exp\!\left(\Delta \ln A^{\text{direct}}\right), \qquad t \geq t_0. \tag{95}$$

This is the "institutional endowment" scenario: Peru immediately operates at German institutional quality but its production structure does not otherwise adjust.

G.3 Step 3: Dynamic Counterfactual (Scenario B)

Better institutions lower the fixed costs of exporting complex products (Section 5). Over time this enables firms to enter new, more complex export sectors, raising the Economic Complexity Index. Scenario B adds this compounding ECI channel to the static level shift.

ECI update. The initial ECI gain ΔECI is estimated from the product-level activation probabilities in Part II (Run 17). For each product whose predicted activation probability rises by at least 0.2 percentage points under the friction shock, and whose product-space distance from the current export basket is below the 75th percentile, we compute the PCI-weighted mean complexity of the new entrants relative to the country's current ECI. This gives the one-time ECI boost that the improved institutional environment makes accessible.

Geometric decay. We assume that in each subsequent year the country can activate a fraction $(1-\delta)$ of the prior year's new ECI additions, with $\delta = 0.15$. What decays is not the ECI level itself, but the *pace* at which new capabilities can be layered on: early gains are largest; later gains are smaller but still strictly positive. The cumulative ECI shift τ years after the reform is therefore

$$\text{ECI}^{\text{cum}}_{\tau} = \Delta\text{ECI} \cdot \sum_{s=0}^{\tau} (1-\delta)^s. \tag{96}$$

Dynamic TFP spell. The additional TFP accumulation relative to Scenario A is

$$\Psi_\tau = \underbrace{\left(\hat{\beta}_{\text{ECI}} + \hat{\beta}^g_{\text{ECI}}\right) \cdot \text{ECI}^{\text{cum}}_{\tau}}_{\text{ECI levels + growth channel}} + \underbrace{\hat{\beta}_{\text{conv}} \cdot \Delta \ln A^{\text{direct}}}_{\text{convergence drag}}, \qquad \tau > 0, \tag{97}$$

where $\hat{\beta}_{\text{ECI}} = 0.184$ is from the TFP levels regression column (3), $\hat{\beta}^{g}_{\text{ECI}} = 0.102$ and $\hat{\beta}_{\text{conv}} = -0.407$ are from the ten-year TFP growth regression column (6). The convergence term is negative: the upward TFP shift places Peru closer to the world frontier, reducing the catch-up growth bonus it would otherwise have enjoyed. The counterfactual GDP per capita path under Scenario B is

$$\hat{y}^{B}_{t} = y_{t} \cdot \exp\big(\Delta \ln A^{\text{direct}} + \Psi_{t-t_0}\big), \qquad t \geq t_0. \tag{98}$$

At $\tau = 0$ the two scenarios coincide; they diverge as the ECI channel compounds over the following decade.

G.4 Key Parameters

Table 17: Parameters used in the counterfactual exercise

Parameter	Value	Source	Role
$\hat{\beta}_{\text{inst}}$	0.032	Part I, back of the envelope exercise.	Inst. quality → TFP level
$\hat{\beta}_{\text{LPI}}$	0.087	Part I, back of the envelope exercise.	Logistics → TFP level
$\hat{\beta}_{\text{ECI}}$	0.184	Part I, Table 1 col. (3)	ECI → TFP level
$\hat{\beta}^{g}_{\text{ECI}}$	0.102	Part I, Table 1 col. (6)	ECI → 10-yr TFP growth
$\hat{\beta}_{\text{conv}}$	−0.407	Part I, Table 1 col. (6)	Convergence drag
δ	0.15	Assumed	ECI decay rate
Δz^{inst}	4.397	WGI, 2008	Institutional gap (PER → DEU)

Note: $\hat{\beta}_{\text{inst}}$ and $\hat{\beta}_{\text{LPI}}$ are drawn from *separate* Part I regressions, each including only one friction composite at a time; no joint specification exists. We run a operate joint specification to compute the values for the counterfactual exercise instead of adding separate coefficients. Factor inputs (K, H) are held at historically observed levels throughout. No general-equilibrium price or factor-market adjustment is made. The exercise should be read as a back-of-the-envelope illustration of orders of magnitude, not as a structural causal estimate.

H Supplementary table referenced in the main text

Table 18: Friction and complexity main effects in recommended FE-decomposition specifications

	Institutional Frictions			Logistical Frictions		
Specification	Friction main effect	Complexity main effect	N	Friction main effect	Complexity main effect	N
Log exports						
Country-unit means, 2005–2015	−1.175*** [0.253]	0.220 [0.379]	128,448	−1.593*** [0.261]	0.265 [0.416]	128,149
Drop country-year and unit-year FE	−1.032*** [0.298]	−0.229*** [0.044]	2,893,412	−0.188 [0.121]	−0.054 [0.038]	2,049,225
Drop country-year FE	−1.025*** [0.298]	–	2,893,412	−0.190 [0.120]	–	2,049,225
Drop unit-year FE	–	−0.236*** [0.045]	3,347,359	–	−0.076* [0.038]	2,325,311
Positive exporter						
Country-unit means, 2005–2015	0.002 [0.017]	0.016 [0.032]	128,448	0.008 [0.017]	0.000 [0.033]	128,149
Drop country-year and unit-year FE	−0.079*** [0.022]	−0.007*** [0.003]	2,893,412	−0.009 [0.010]	−0.002 [0.002]	2,049,225
Drop country-year FE	−0.079*** [0.022]	–	2,893,412	−0.009 [0.010]	–	2,049,225
Drop unit-year FE	–	−0.007** [0.003]	3,347,359	–	−0.003 [0.002]	2,325,311

Notes: Cells report main-effect coefficients from the recommended FE-decomposition specifications. Blank main effects in the source table are shown as dashes because the corresponding fixed effects absorb the main effect. Standard errors are reported in square brackets. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

I Additional Diagnostics for Part II

I.1 Sample diagnostics

Since sample inclusion varies only with the availability of the friction proxy, not with the choice of export outcome, Table 19 collapses the diagnostics across outcome definitions and reports the country-year weighted sample composition by friction-data block.

Table 19: Sample coverage and income tilt by friction-data family

Friction-data family	Country-year coverage	Mean GDPpc comparison included vs. potential	Interpretation
WDI: access to electricity	94.9%	21.98 vs. 21.70 gap: +0.28, +1.3%	Near-complete coverage; the included country-years look very similar to the full potential sample.
WGI: institutional quality	85.3–85.8%	22.20–22.29 vs. 21.70 gap: +0.51–+0.59, +2.3–+2.7%	Broad coverage, with only a mild tilt towards richer country-years.
WDI: electric power losses	76.6%	22.98 vs. 21.70 gap: +1.28, +5.9%	Moderate coverage loss; the estimation sample is somewhat richer than the potential sample.
LPI: logistics performance	53.8%	23.63 vs. 21.70 gap: +1.93, +8.9%	Main sample-coverage bottleneck; LPI availability induces the strongest income tilt.

Notes: Statistics are country-year weighted. GDP per capita means are reported in thousands of dollars. The potential-sample GDPpc mean is 21.70 throughout. The gap is the included-sample mean minus the potential-sample mean; the percentage in parentheses reports the gap relative to the potential-sample mean. Outcome definitions are collapsed because sample inclusion is determined by friction-proxy availability. For grouped friction-data families, ranges report the minimum and maximum across proxies in the group.